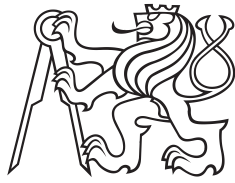


Master Thesis



Czech  
Technical  
University  
in Prague

**F3**

Faculty of Electrical Engineering

## Numerical Optimal Control for Minimum Lap Time

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Supervisor: doc. Ing. Zdeněk Hurák, Ph.D.  
May 2026



## Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

## Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 22. May 2026

## Abstract

Thesis researches and implements ways of solving minimum maneuvering time problem (lap time simulation) for road vehicles. Thesis divides this problem into smaller subproblems, for which there are already mature software methods available. Subproblems are: racer car modelling, transcription of optimal control problem into NLP, solving of NLP, solving sets of linear equations and sensitivity analysis. Thesis implements these methods and benchmarks their performance. x

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**Keywords:** word, key

**Supervisor:** doc. Ing. Zdeněk Hurák, Ph.D.

## Abstrakt

Diplomova práca skúma metódy riešenia minimum maneuvering time problému pre cestne vozidla a implementuje ich. Práca delí tento problém na menšie podproblémy: vytvorenie modelu vozidla, zápis problému optimálneho riadenia, riešenie nelineárneho programu, riešenie systému lineárnych rovníc a citlivosťná analýza. Pre tieto podproblémy existuje mnoho rôznych možností riešenia, práca tieto možnosti vysvetľuje a porovnáva. V práci sú implementované metódy riešenia podproblémov ktoré som považoval za najužitočnejšie. Práca obsahuje aj benchmark vybraných možností riešenia.

**Klíčová slova:** NLP, laptime simulation, ipopt, skopirovat z bakalarky

**Překlad názvu:** Metody numerického optimálního řízení pro minimalizaci času na kolo — Cesta do tajů kdovíčeho

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# Chapter 1

## Introduction

### 1.1 Motivation

Topic of this thesis is motivated, by my work in Formula Student team eForce Prague Formula, where I worked on real-time vehicle dynamics control software. There I got intricately exposed to problems faced by racing teams, such as should we increase the aerodynamic downforce at the cost of higher drag? Should we put accumulator in the middle of vehicle, to decrease moment of inertia? Should the aerodynamics be designed for straight line drive, or for steady state turning? How to use limited energy from accumulator more effectively, how faster can we go if we made better power limiting algorithm? How large the accumulator should be? Which torque allocation algorithm should be used? How to design the suspension kinematics to maximize the tire utilization? Is rear wheel steering worth the increased weight? All of these problems can be analyzed by using purpose built tool, good engineering practices or using pen and paper with heavily simplified model. Formula student season is 10 months for development and construction of the vehicle, 2 months for competition. That puts enormous pressure on deciding on vehicle concept fast. In my opinion what formula student teams need is a tool to quickly evaluate different car concepts. As my bachelor thesis I created a prototype of such tool, goals of this thesis come from computational and user experience from my previous tool.

### 1.2 Goals

- Primary goal of this thesis is to investigate available methods for numerical optimal control with respect to their suitability for planning an optimal trajectory of a vehicle aiming to achieve the minimum lap time on a racing circuit. To do this for various vehicle concepts, thesis has to present a vehicle modelling framework.
- Benchmark performance of available numerical methods
- Thesis should present a tool for convenient vehicle modelling, and vehicle concept evaluation.

- Support vehicle parameter optimization, using any vehicle parameter as control input and sensitivity analysis of vehicle parameters.
- Goal is not precise vehicle modeling and validation, goal is to create a framework to perform vehicle concept analyses easily.

### 1.3 Structure of thesis

First part of thesis with vehicle modelling in the context of optimal control. A universal model structure inspired by [16] is presented. Then the theory of optimal control idem spat ja neviem co pisat uz.

1. ODE models of cars
2. Solving methods: time marching forward, backward pass, NLP,
3. Collocation (p, h, hp refinement), integral + differential form (Gauss-Radau, Legendre, Lobatto)
4. Implementation overview, Algorithmic differentiation, JUMP/Examples/InfiniteOpt + Julia, Python/MATLAB, CasADi, interior point solvers
5. Actual implementation and results
6. Conclusion

### 1.4 Literature overview/ survey of laptime simulators

Lap-time prediction dates back to the 1950s. Computers became common in the 1970s and quasi-steady-state (QSS) methods became the standard approach for MLTS. QSS is still what most race engineers mean by “lap-time simulation”. In QSS the vehicle is a point mass on a fixed racing line, with lateral and longitudinal accelerations bounded by a steady-state performance envelope [52]. Transient effects such as yaw acceleration are ignored. MLTS was first formulated as an optimal control problem in the late 1980s [28]. A survey of current methods is given in [44]. The work of Perantoni and Limebeer [50] is the main reference: it uses a track-fixed coordinate system, switches the independent variable from time to travelled distance, and applies the formulation to a twin-track Formula 1 model with some vehicle parameters as decision variables. Parts of this thesis build on it. MLTS is still an active research area because better predictions give measurable setup gains, but higher fidelity models cost more solver time. The authors of [2] use sequential convex programming to cut compute cost without losing accuracy.

A very popular laptime simulator is OptimumLap from OptimumG [3], It is a free commercial tool advertised to achieve 10% accuracy with real vehicles. It uses Quasi Steady State methods (QSS) This is the standard tool

for laptime simulations. Open source implementation of the same method [45], My implementation of this method in chapter ?? . QSS methods are very fast, but the models for which they can be used is equally very simple.

Popular open source C++ implementation of laptime simulator using nonlinear programming and automatic differentiaoton[43]. It uses IPOPT and CppAD[7]. Vehicle model is 6DOF twintrack. Other implementations are work from austria or germany[29] class project from Georgia institute of technology [1]





# **Part I**

## **Theory**



## Chapter 2

### Vehicle and Track modelling framework

Goal of the thesis is to select a sufficiently accurate yet computationally tractable model of the vehicle. There is not one model of vehicle that would fit all requirements of race engineers. For each problem, a different complexity of the vehicle model is sufficient, e.g. for analysis of power limit algorithm, the lateral dynamics of the vehicle can be heavily simplified and simple transcription and solving methods yield sufficient accuracy within a fraction of a second, this is described in ???. Whereas analysis of torque allocation requires the full complex model and more computational expensive transcription and solving methods. Therefore, a vehicle modeling framework is needed to satisfy the needs of different types of analyses.

#### 2.1 Model like in MSD

The structure of the vehicle model has to allow extending/changing components, so that different vehicle concepts can be simulated, with minimal coding. Book in [16] describes modelling framework that allows to create car components as modules with flow and effort interface. This common interface then heavily simplifies process of creating a car model, as modules can be reused and chained together easily. ADD DESCRIPTION OF THE BROWN GRAPH THEORY HERE. velocity goes one direction and forces go the other direction, must give better description.

Vehicle components are chassis, wheel pivot, tire, aerodynamics, suspension, motor

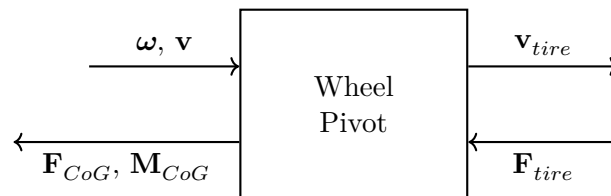
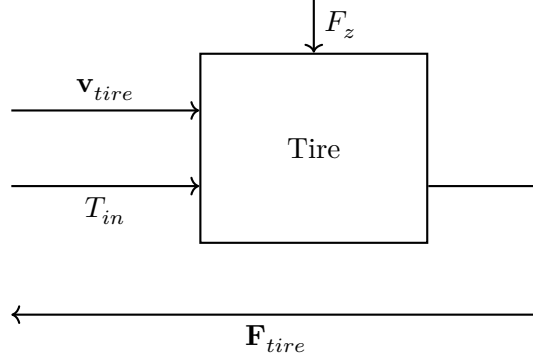


Figure 2.1: Wheel Pivot

## 2.2 Tire - road interface modelling

Job of the tire model in this thesis is to calculate forces acting on tire from the relative velocity and angle between tire and road. Input of general tire model is 3 speeds and 3 angles, output is 3 forces.



**Figure 2.2:** Tire component

Most forces that govern vehicle movement are transferred by the tires. The tire is considered to be the most important system of a vehicle and it is notoriously difficult to model [46]. Thus tires need to be modeled with great caution in mind. Not only are there many parameters, such as tire wear, tire temperature, road roughness, road wetness, tire pressure, etc., that affect the resultant forces, they are also constantly changing during the lap. Therefore they are difficult to measure and model precisely. TOTO SOM SKOPIROVAL Z BAKALRKY, V CLANKOCH SOM VIDEL DOSLOVNE KOPIROVANIE SEBA, MOZEM TO ROBIT JA? Implementation of specific models is conver in part III Implementation. [18].

## 2.3 Chassis model

point mass with definition of wheel pivot points.

$$\mathbf{M}_{CoG,i} = \mathbf{r}_i \times \mathbf{F}_i, \quad \mathbf{F}_i = \begin{bmatrix} F_{x_i} \\ F_{y_i} \\ 0 \end{bmatrix}. \quad (2.1)$$

$$\mathbf{F}_{CoG} = \sum_i \mathbf{F}_i + \begin{bmatrix} F_{\text{drag}} \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{M}_{CoG} = \sum_i \mathbf{M}_{CoG,i}. \quad (2.2)$$

$$\dot{\mathbf{v}} = \frac{\mathbf{F}_{CoG}}{m} - \boldsymbol{\omega} \times \mathbf{v}, \quad \dot{\boldsymbol{\omega}} = \frac{\mathbf{M}_{CoG}}{I}. \quad (2.3)$$

$$\mathbf{x} = \begin{bmatrix} v_x \\ v_y \\ \psi \\ \omega_z \end{bmatrix}, \quad \dot{\mathbf{x}} = \begin{bmatrix} \dot{v}_x \\ \dot{v}_y \\ \dot{\omega}_z \\ \dot{\omega}_z \end{bmatrix}. \quad (2.4)$$



## 2.4 Wheel Pivot model

Wheel pivot point is modeled as transformer. It defines the transformation of CoG velocities into tire frame and then transforms tire forces and torque into CoG frame. Angular velocity and position vectors are formed from yaw rate  $\dot{\psi}$  and the longitudinal distance  $l_i$  of the axle from the center of gravity:

$$\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix}, \quad \mathbf{r}_i = \begin{bmatrix} l_i \\ b/2 \\ 0 \end{bmatrix}, \quad (2.5)$$

where  $b$  is the track width. Velocity at each tire pivot point is:

$$\mathbf{v}_{p_i} = \mathbf{v} + \boldsymbol{\omega} \times \mathbf{r}_i. \quad (2.6)$$

Velocities are then rotated from the pivot frame into the tire frame using the steering angle  $\delta_i$ :

$$R_t = \begin{bmatrix} \cos \delta_i & -\sin \delta_i & 0 \\ \sin \delta_i & \cos \delta_i & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{v}_{t_i} = R_t \mathbf{v}_{p_i}. \quad (2.7)$$

Tire forces  $F_{tx}$  and  $F_{ty}$  are computed from  $\mathbf{v}_{t_i}$  by the tire model. Forces are then rotated back to the pivot frame:

$$\begin{bmatrix} F_{x_i} \\ F_{y_i} \end{bmatrix} = \begin{bmatrix} \cos \delta_i & \sin \delta_i \\ -\sin \delta_i & \cos \delta_i \end{bmatrix} \begin{bmatrix} F_{tx} \\ F_{ty} \end{bmatrix}. \quad (2.8)$$

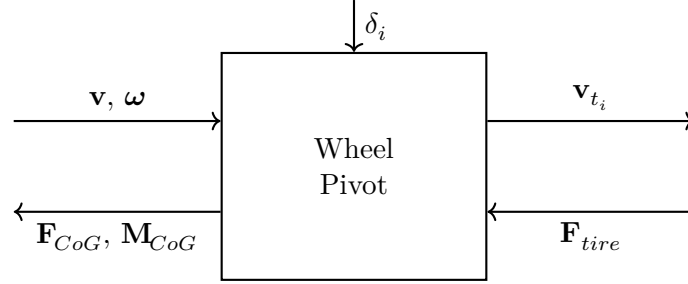


Figure 2.3: Wheel Pivot component

## 2.5 Aerodynamics model

Aerodynamic model is modelled as nonlinear resistance element, mapping velocity into forces. Drag and downforce scale with the square of longitudinal velocity:

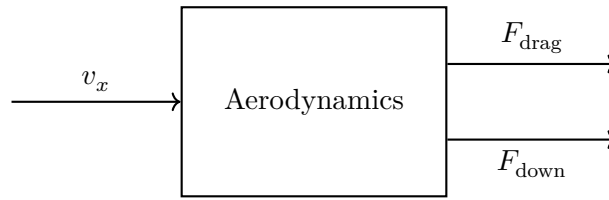
$$F_{\text{drag}} = \frac{1}{2} \rho C_D v_x^2, \quad F_{\text{lift}} = \frac{1}{2} \rho C_L v_x^2. \quad (2.9)$$

The downforce is split between front and rear axles by the center of pressure ratio  $\text{CoP} \in [0, 1]$ , where  $\text{CoP} = 0.5$  corresponds to a balanced distribution.

**Table 2.1:** Parameters of the aerodynamic model.

| Symbol | Unit              | Description                                |
|--------|-------------------|--------------------------------------------|
| $C_L$  | —                 | Lift coefficient (negative = downforce)    |
| $C_D$  | —                 | Drag coefficient                           |
| CoP    | —                 | Center of pressure (front axle share, 0–1) |
| $\rho$ | kg/m <sup>3</sup> | Air density                                |

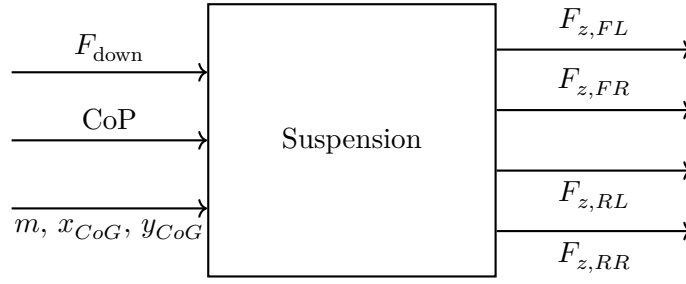
Air density  $\rho$  is taken as 1.225 kg/m<sup>3</sup> at sea level. Coefficients  $C_L$ ,  $C_D$  and CoP used per vehicle model are listed in Table 2.1.

**Figure 2.4:** Aerodynamics component

This model is sufficient when vehicle is driving relatively straight, that means lateral acceleration is roughly zero. With high lateral acceleration this model is no longer valid, because air speeds are different on left and right part of car and also the car starts rolling- one side of vehicle is closer than the other, creating imbalance of lift forces.

## 2.6 Suspension model

Job of suspension in car is to maintain tire contact, distribute the load from lateral load transfer to better utilize tires and use kinematics to make use of camber thrust. Commonly suspension of car consists of spring, damper and the mounting with tie rods. In this thesis Suspension model had to be significantly simplified. Modelling each wheel with a damper, spring and its own kinematics is beyond the scope of this thesis. However, the structure of allows it, i just didnt have enough time to implement it. this quarter car model would be very beneficial toward the goal of scalability, now without it the model scalability is limited as divigin force between more than two wheel on axis is staticky nedoruceny problem.



**Figure 2.5:** Suspension component

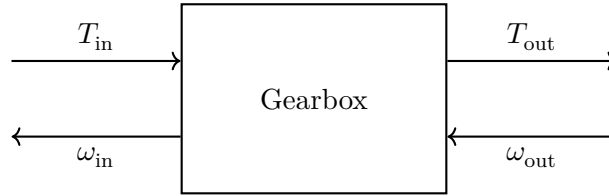
## 2.7 Gearbox model

gearbox is the textbook example of transformer and resistive element.

$$T_{\text{out}} = \eta i T_{\text{in}}, \quad \omega_{\text{out}} = \frac{\omega_{\text{in}}}{i}. \quad (2.10)$$

**Table 2.2:** Parameters of the gearbox model.

| Symbol | Unit | Description                        |
|--------|------|------------------------------------|
| $i$    | —    | Gear ratio (input to output speed) |
| $\eta$ | —    | Mechanical efficiency              |



**Figure 2.6:** Gearbox component

## 2.8 Accumulator model

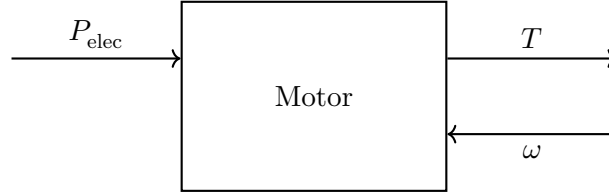
Accumulator is modelled as capacitor from brown, with additional constraints on maximum and minimum power

## 2.9 Motor model

$$P_{\text{mech}} = T \omega, \quad P_{\text{elec}} = \max\left(\frac{P_{\text{mech}}}{\eta}, P_{\text{mech}} \eta\right). \quad (2.11)$$

**Table 2.3:** Parameters of the motor model.

| Symbol            | Unit | Description                          |
|-------------------|------|--------------------------------------|
| $T$               | Nm   | Shaft torque                         |
| $\eta$            | —    | Motor + inverter efficiency          |
| $P_{\text{mech}}$ | W    | Mechanical shaft power               |
| $P_{\text{elec}}$ | W    | Electrical port power (battery side) |



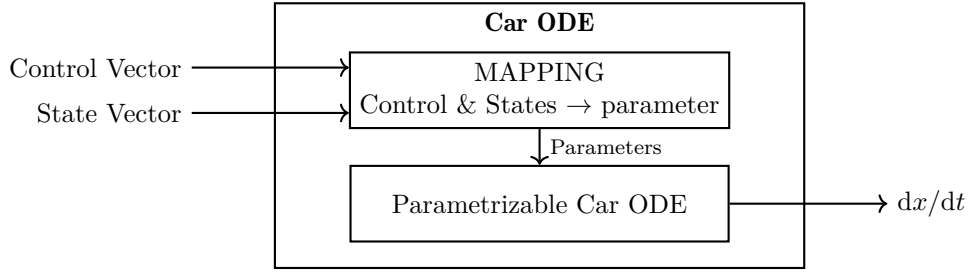
**Figure 2.7:** Motor component: electrical power  $P_{\text{elec}}$  enters from the battery side, shaft angular velocity  $\omega$  is imposed by the drivetrain, output is shaft torque  $T$ .

## 2.10 Car models

These models must work when used in NLP transcription for solving boundary value problem, but they also must work for initial value problem, when the model is initialized or verified for transcription error. Some toolboxes can automatically create constraints from ODE models[61]. In my implementation I have and if statement that switches between IVP and BVP variant of models equations, major disadvantage of this approach is that the equations for both variants must be effectively the same and finding errors is difficult in this.

### 2.10.1 Generic car model

one of the goals is calability, therefore any car Parameter can be assigned as state, control, static parameter, decision variable or sensitivity analysis variable. This allows for simple RELAXATION? rozvolnenie? of DOF allowing to decide on one line of code which parameters should be controlled. This also allows for finding the optimal parameters of vehicle, such as brake bias, aerodynamics balance, gear ratio. When parameter is chosen to be sensitivity analysis variable, impact of changing this variable is cheaply computed after the optimisation is finished, see Section 3.9.



**Figure 2.8:** Generic car model structure. Control and state vectors are mapped to car parameters, which are then used by the parametrizable car ODE to compute derivatives.

### 2.10.2 Mass Point Quasi Steady State model

Simple and useful car model can be made using these equations

$$F_z = \frac{1}{2} \rho C_L v_x^2, \quad (2.12)$$

$$F_y = m v_x^2 c, \quad (2.13)$$

$$F_{x,\text{pwr}} = \frac{P_{\max}}{v_x}, \quad (2.14)$$

$$F_{x,\max}^2 = \max\left([ (F_z + mg)\mu ]^2 - F_y^2, 0\right), \quad (2.15)$$

$$F_x^2 + F_y^2 \leq [(F_z + mg)\mu]^2, \quad (2.16)$$

$$|F_x| \leq \min(F_{\text{mot},\max}, F_{x,\text{pwr}}), \quad (2.17)$$

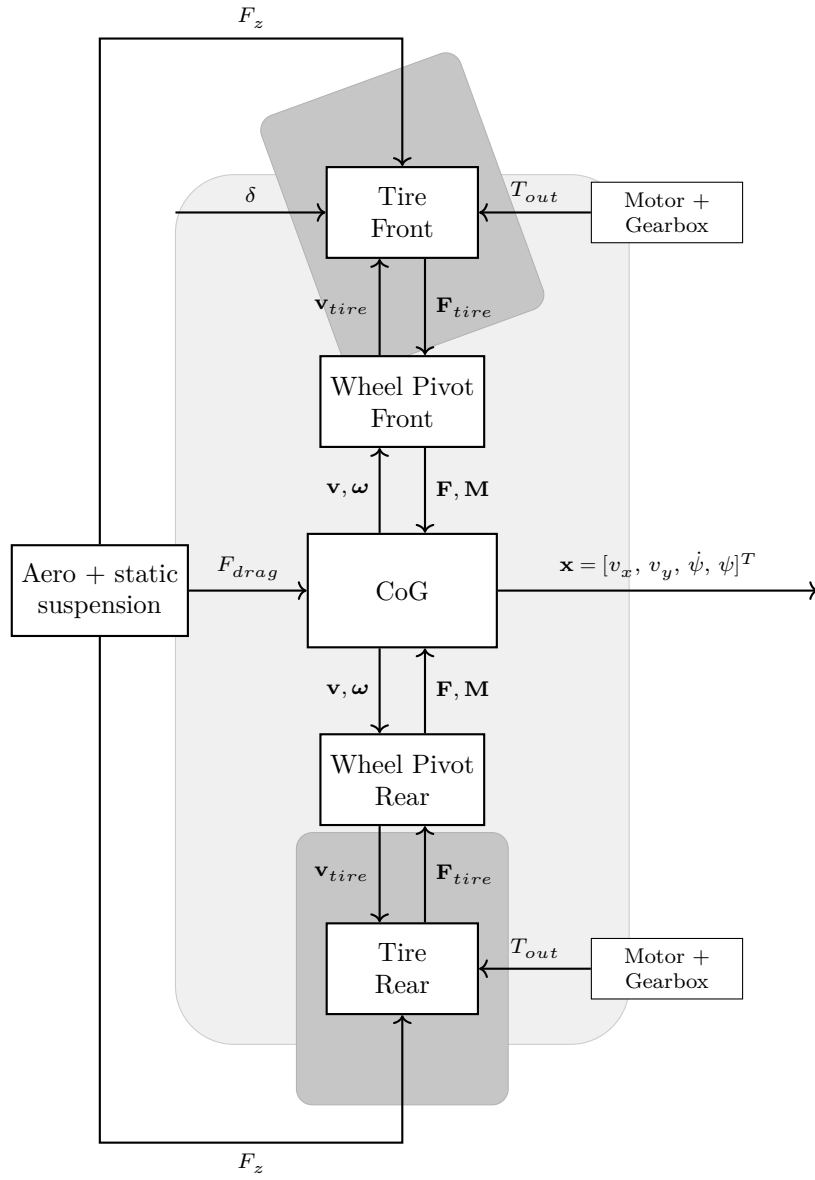
$$\dot{v}_x = \frac{F_x - \frac{1}{2} \rho C_D v_x^2}{m}. \quad (2.18)$$

Where  $\rho$  is air density  $C_L$  is lift coefficient  $v_x$  is longitudinal speed,  $m$  is mass,  $c$  is curvature of track in current point,  $P_{\max}$  is maximum power of motor,  $\mu$  is friction coefficient,  $C_D$  is drag coefficient.

This model has only one state  $v_x$  which makes it possible to use specialized solvers. Variations of this model, where for example  $P_{\max}$  is a lookup table, the grip does not scale linearly with downforce etc. are commonly used in motor-sport.

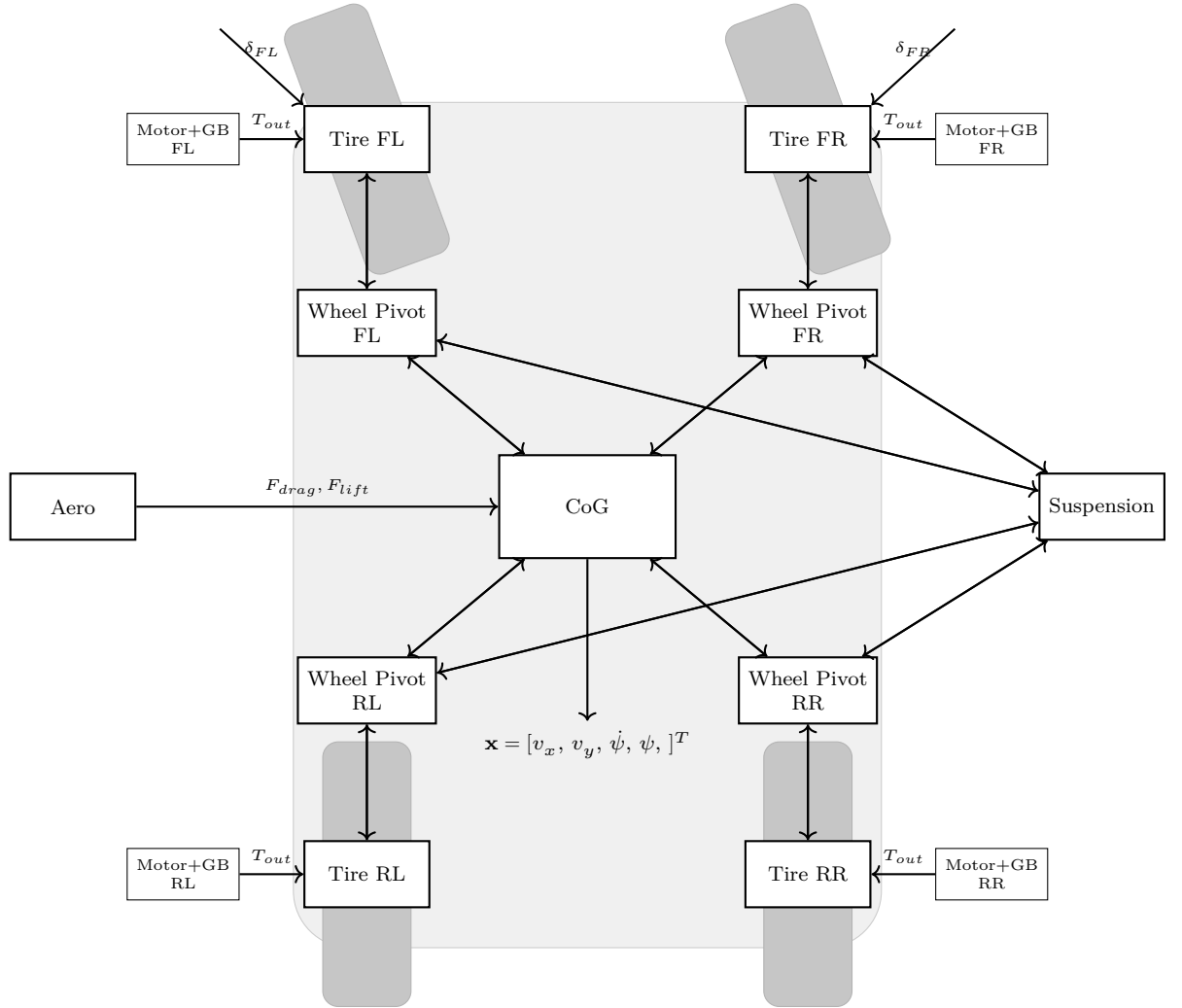
### 2.10.3 Single track

A more complex model and useful model is single-track. In this model the suspension effects are neglected. Singletrack is still considered to be simplified model and is used most commonly when computation limits are reached.



**Figure 2.9:** Single-track (bicycle) model computational graph. Velocity flows from CoG through wheel assemblies to tires, forces flow back. Motors provide torque through gearboxes, aerodynamics provides drag and normal forces.

### 2.10.4 Twintrack model



**Figure 2.10:** Twin-track model. Velocity flows from CoG through wheel pivots to tires; tire forces flow back. Motors deliver torque through gearboxes. Aerodynamics acts on the chassis; suspension exchanges loads and displacements with each wheel pivot. The chassis state  $\mathbf{x}$  is exported below the CoG.

## 2.11 Track modeling

Track is sampled at discrete arc length nodes  $s_i$  and stored as a set of fields evaluated at each node:

- centerline coordinates  $x(s), y(s)$ ,
- heading  $\theta(s)$ ,
- curvature  $\kappa(s)$ ,

- left and right track widths  $w_L(s)$ ,  $w_R(s)$ ,
- air density  $\rho(s)$ ,
- friction coefficient  $\mu(s)$ ,
- inclination  $\phi(s)$ .

Each field is linearly interpolated between the segments. Obtaining suitable track for optimization from GPS is non-trivial, simple methods lead to noisy data, which then corrupts the optimization result. Therefore, smoothing method described in [50] has been used.

## ■ 2.12 Change of independent variable, time to path

When differential equations for vehicle movement are stated. For use in lap time simulation it is beneficial to describe the car states not in time but in distance traveled. This introduces connection with location on track and reduces number of state variables by one. This transformation is taken from [50].

$$\varepsilon = \psi - \theta(s) \quad (2.19)$$

where  $\psi$  is car heading,  $\theta(s)$  is heading of track at distance  $s$ ,

$$S_f = \frac{dt}{ds} = \frac{1 - n C(s)}{v_x \cos \varepsilon - v_y \sin \varepsilon}, \quad (2.20)$$

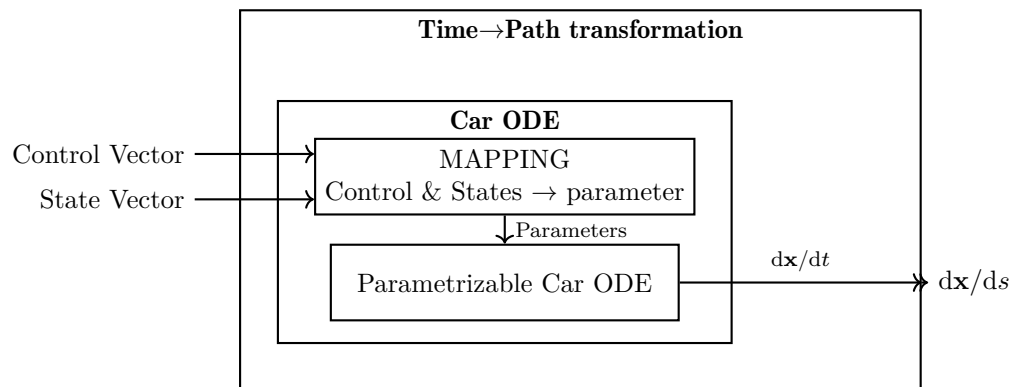
$n$  is vehicles distance from centerline,  $C(s)$  is curvature of track at distance  $s$ . The change in  $n$  is defined as:

$$\frac{dn}{dt} = v_x \sin \varepsilon + v_y \cos \varepsilon. \quad (2.21)$$

Then the model can be transformed from time as independent variable to distance travelled along centerlines as independent variable by equation:

$$\frac{d}{ds} \begin{pmatrix} v_x \\ v_y \\ \psi \\ \omega \\ n \\ t \end{pmatrix} = S_f \begin{pmatrix} \dot{v}_x \\ \dot{v}_y \\ \dot{\psi} \\ \dot{\omega} \\ \dot{n} \\ 1 \end{pmatrix}. \quad (2.22)$$





**Figure 2.11:** Generic car model wrapped in a time-to-path transformation layer

Now the vehicle is modeled and placed on the track. Next step is to drive it as fast as possible.



## Chapter 3

### The Optimal Control Problem

In previous chapter I have described how to build differential equations that govern movement of a vehicle. This chapter describes two approaches of driving that vehicle optimally, first by using classical initial value problem/time marching formulation, then reformulating it to boundary value problem.

#### 3.1 Minimum maneuvering time problem as initial value problem

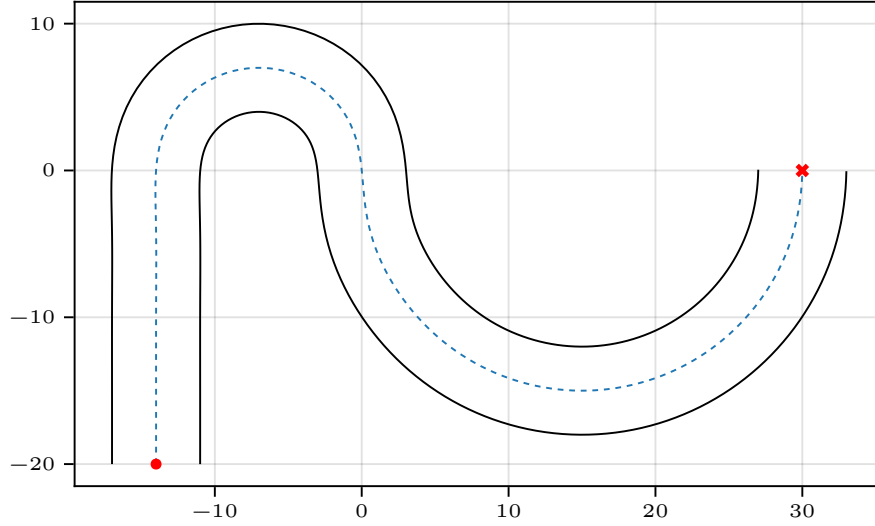
The most common method for solving an ODE is as initial value problem. An industry standard high fidelity multi-body time marching solver for vehicles is Car Maker from IPG [?]. It is great for driving simulation, testing and development of advanced driver assistant systems, modeling the vehicles and racetrack modeling, it can be used for lap-time simulation, but with caveats. In time marching solvers a separate controller is required, which becomes part of the ODE. This approach gives a solution to the problem of how to control car, so that it completes a lap around a circuit. However, this solution is almost certainly not optimal to time or any meaningful objective. Another issue is that if the vehicle's parameters change and the controller does not, the resulting time can be exactly same, because controller cannot take advantage of different vehicle's properties. Sensitivity analysis and lap time can be obtained but it is skewed by the chosen controller. The controller can be tuned for specific vehicle, but that will still not guarantee optimal control. Using time marching is not the best for sensitivity analysis and lap time simulation of a complex model.

When the vehicles model is simplified to Quasi Steady State model, time marching simulation can be used.

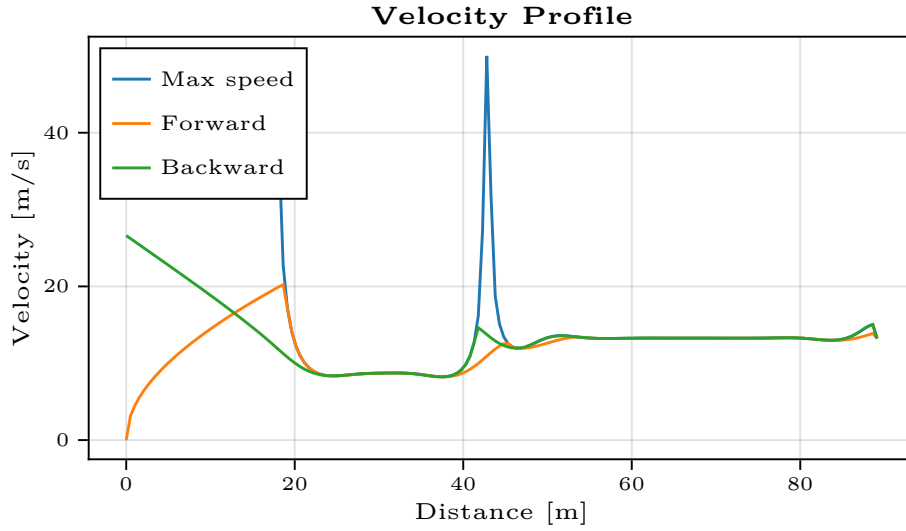
##### 3.1.1 Quasi Steady state lap time simulation

Following section describes commonly used algorithm which uses only initial value solvers, to compute the minimum maneuvering time. For quasi steady state model 2.18 it is possible to compute optimal speed profile ked dal som rovanke rovnice do nlp a dostal som rovnaky vysledok ako z toho algoritmu, takže aspon pre tuto trat to asi bude optimum speed profile using an algorithm

well known among racing engineers. Figure 3.2 shows the speed profile obtained from this algorithm, on track from Figure 3.1, start is dot and cross is finish.



**Figure 3.1:** Test track used for the forward/backward pass example.



**Figure 3.2:** Speed profile from forward and backward passes with Quasi Steady state model

Algorithm works in following manner:

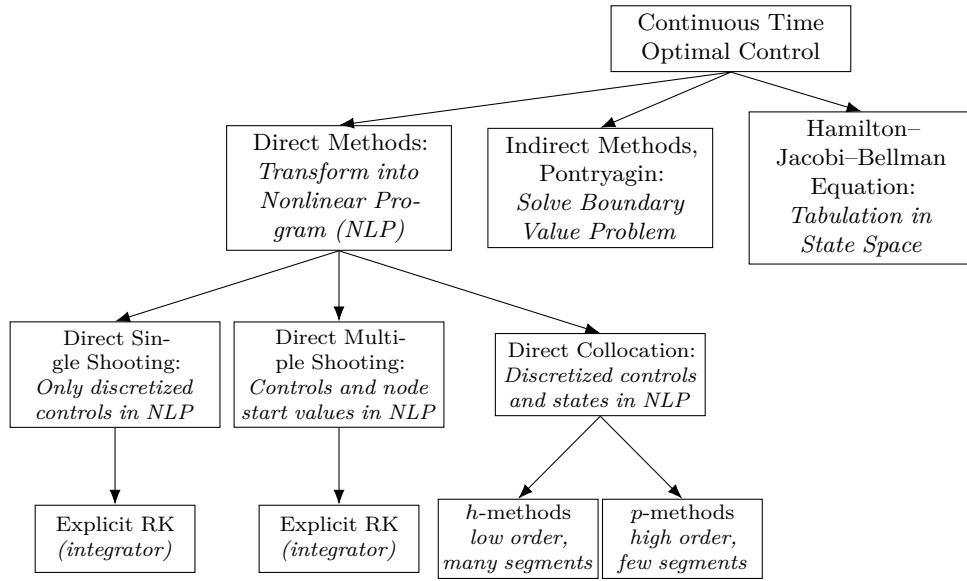
1. Maximum speed Pass: From the vehicle model and curvature along track, compute the maximum speed, blue line on Figure 3.2

2. Forward pass: consider only forward acceleration limits of the vehicle (braking force is unlimited), run the vehicle from start to finish along the trajectory, applying maximum longitudinal acceleration and respecting max speed limit, orange line on Figure 3.2
3. Consider only braking acceleration limits of the vehicle (acceleration force is unlimited), run the vehicle *backwards, from finish to start* along the trajectory, applying maximum braking acceleration (braking now increases speed as the simulation is flipped) and respecting max speed limit
4. Take minimum speeds of forward and backward pass and this gives the minimum time speed profile

More complete description of this algorithm is in [?] This is the classic approach to minimum maneuvering time problem, the issue is that the trajectory of vehicle on track is fixed and cannot be optimized, the vehicle model must be heavily simplified, it does not work with any integral constraints like accumulator energy. However, the computation time can be fraction of a second for the entire lap. Tools using this method have proven to be extremely useful and many different analyses can be performed by leveraging its computational speed, for example varying the friction coefficient at on each corner, a corners importance on lap time can be calculated [?].

## 3.2 Transcription methods

Previous section described automotive specific methods of solving minimum maneuvering time problem, this section explains methods of transcribing the problem into the language of optimal control problem. Transcription methods rewrite continuous (infinite-dimensional) OCP into finite-dimensional NLP. That means representing the continuous trajectory of a state by values sampled at discrete points. The terminology of transcription methods can be difficult to understand, as it is complex topic and often articles present conflicting information [40]. Website [40] and other work from Matthew Kelly [37, 35, 38] offers a very digestible introduction into the topic. This thesis focuses only on direct methods. Classification of methods in this thesis is based on [40] Figure 3.3 shows how transcription methods are commonly classified.



**Figure 3.3:** Tree of approaches to continuous-time optimal control, adapted from [?], with the direct-methods branch expanded.

### 3.2.1 Shooting methods

Shooting methods get their name from aiming artillery canon analogy, with given powder load canon and target location find the aiming angle to hit the target. In depth explanation with animation and example code is here [36] This is classical boundary value problem. Methods shoot, miss, adjust aim, shoot again and iterate until target is hit. These methods are solving boundary value problem using an IVP solver. More precisely,

1. Guess the controls
2. Compute the solution using an initial value problem solver
3. Compare the simulated value at the end of interval from 2. with the boundary value(targets location)
4. Based on error update the controls and go to 2.

Shooting methods are split into single and multiple shooting. They differ in way they handle state discretisation. Single shooting discretizes the state only at start and end of interval and computes the trajectory in one shot. Whereas multiple shooting discretizes the states along the trajectory and computes multiple shots, this can be also parallelized on a CPU [22]. Shooting methods are progressively being displaced by collocation methods [51]

### 3.2.2 Nonlinear Programming with Runge Kutta constraints

These methods are not using initial value solvers to solve boundary value problem, instead they use nonlinear programming. Most straightforward

method to transcribe continuous optimal control problem into non-linear program, is to use explicit Runge Kutta solving methods from initial value problem as equality constraints on the dynamics of a vehicle. Using Euler method, the constraints for the dynamics of vehicle are:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + h \cdot f(\mathbf{x}_k, \mathbf{u}_k), \quad (3.1)$$

where  $f$  denotes the vehicle dynamics,  $h$  the step size, and  $\mathbf{x}$  the state vector. This works, but it may be the worst possible transcription method, due to high error and instability of Euler method [4]. Problems of Euler method encountered when solving initial value problems also translate to nonlinear programming. Implicit RK methods can be used too, In Initial value problems they are used less often, as they require iterative solving. Optimal control problem is Boundary value problem and not initial value problem, therefore:

...for a boundary value problem ...any scheme becomes effectively implicit. Thus, the distinction between explicit and implicit initial value schemes becomes less important in the BVP context.

[12, p. 69]

For boundary value problem it makes no sense to use explicit Runge Kutta methods, as they have inferior stability compared to implicit Runge Kutta methods.[51, p. 4]. Implicit Runge Kutta methods, which are commonly used are from LobattoIIIA family, but any other method can be used. Book [15] uses only these.

### 3.2.3 Collocation methods

The concept of collocation is old and universal in numerical analysis ...For ordinary differential equations it consists in searching for a polynomial of degree  $s$  whose derivative coincides ("co-locates") at  $s$  given points with the vectorfield of the differential equation ...[5, p. 224]

The article [37] is an in depth explanation of inner working and implementation of direct collocation methods, theory here is based on it and on [23]. Collocation methods are a subset of implicit Runge-Kutta methods. The distinguishing property is that the solution of a boundary value problem is approximated by a piecewise polynomial in the independent variable (distance along the centerline). On each interval the polynomial is fixed by requiring it to satisfy the ODE exactly at  $N$  chosen points the collocation nodes.

There are numerous ways to construct a collocation method, in optimal control the classic methods are those based on roots Legendre or Chebyshev type I polynomials, Lagrange polynomials [?, 24] and Gauss quadrature rule. In this thesis only Legendre and Gauss quadrature based collocation methods are covered. Figure 3.4 depicts multiple ways of constructing collocation methods.

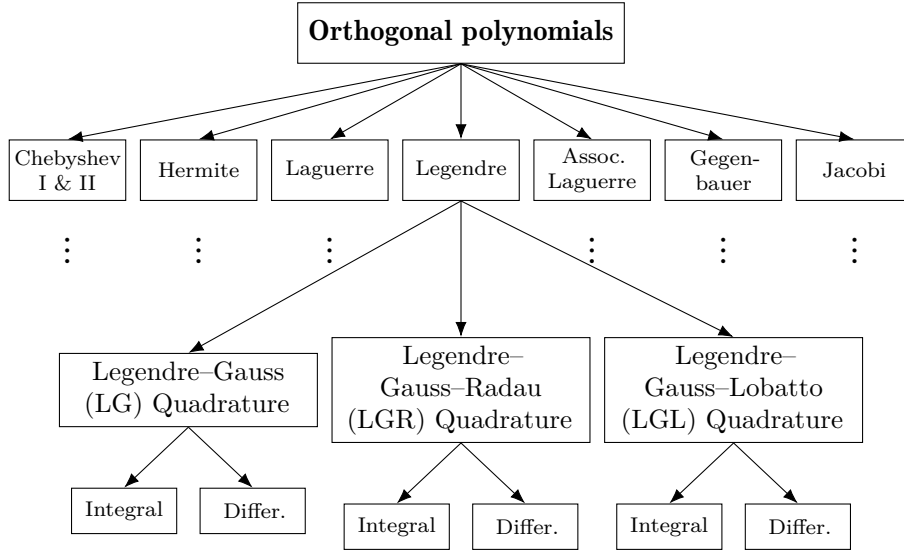


Figure 3.4: Collocation transcription methods

### ■ Lagrange polynomial

First part of creating a collocation method is creating an interpolating polynomial basis,

$$L_i(\tau) = \prod_{\substack{j=0 \\ j \neq i}}^N \frac{\tau - \tau_j}{\tau_i - \tau_j}, \quad i = 0, \dots, N. \quad (3.2)$$

Where  $\tau$  is the point to be interpolated and  $\tau_j$  and  $\tau_i$  are node positions, in order to get interpolating polynomial  $L(\tau)$ , values at nodes  $x_j$  are multiplied by corresponding Lagrange base and summed

$$L(\tau) = \sum_{i=0}^N f_i L_i(\tau). \quad (3.3)$$

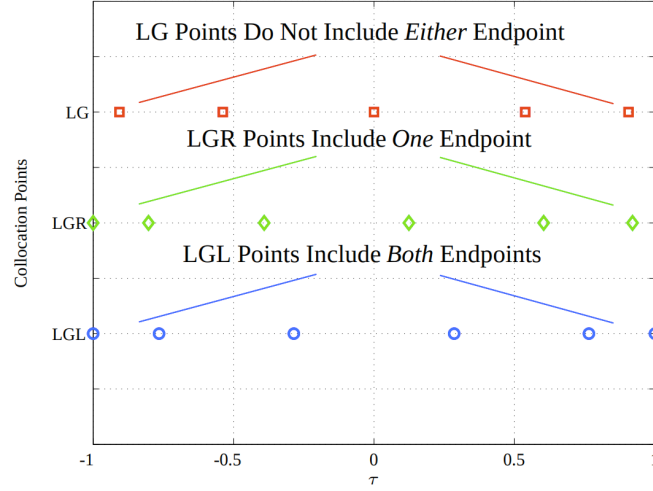
Now it is possible to interpolate a state between the discretized nodes, Note that the node locations for Legendre polynomial are arbitrary. In the actual implementation, calculation of polynomial basis from definition 3.2 is numerically impractical, therefore barycentric form from [13] has been used as implementation of these methods. The next step is to compute definite integral using values from function values at nodes.

### ■ Gaussian Quadrature rule

Quadrature rule is the approximation of definite integral of a function by a sum of function values at given nodes. In optimal control context Gaussian quadrature is commonly used

$$\int_{-1}^1 f(\tau) d\tau \approx \sum_{i=0}^n w_i f_i, \quad (3.4)$$





**Figure 3.5:** Schematic Showing the Differences Between LGL, LGR, and LG Collocation Points, taken from [23]

where  $w$  are Gaussian quadrature weights. A useful property of Gaussian quadrature of order  $n$  is that it approximates the integral to machine precision whenever the integrated function is a polynomial of degree at most  $2n - 1$ . Node selection for Gauss quadrature is *not arbitrary*, in collocation methods the node positions are based on roots of Legendre polynomial. Node positions specify the type of collocation method and are creating in following manner. Let  $P_N(\tau)$  denote the Legendre polynomial of degree  $N$ . Three node families are commonly used [23]:

- Legendre-Gauss: roots of  $P_N(\tau)$ , includes no endpoints.
- Legendre-Gauss-Radau: roots of  $P_{N-1}(\tau) + P_N(\tau)$ , includes one endpoint.
- Legendre-Gauss-Lobatto: roots of  $\dot{P}_{N-1}(\tau)$ , includes both endpoints.

### 3.3 Collocation at Legendre-Gauss points

Legendre-Gauss collocation methods uses nodes strictly inside the interval for approximation using polynomial. With  $n$  nodes, it is possible to approximate functions up to degree  $2n - 1$  with machine precision. States are approximated by series

$$x_j^N(\tau) = \sum_{i=0}^N x_{ij} L_i(\tau), \quad (3.5)$$

where  $j$  is the number of the state. Then the series is differentiated using barycentric form from [13, p.513] and evaluated at  $k$ -th collocation point  $\tau_k$

$$\dot{x}_j^N(\tau_k) = \sum_{i=0}^N x_{ij} \dot{L}_i(\tau_k) = \sum_{i=0}^N D_{ki} x_{ij}, \quad D_{ki} = \dot{L}_i(\tau_k). \quad (3.6)$$

Matrix  $\mathbf{D}$  is called *Gauss Pseudospectral differentiation matrix*. In matrix notation can be written

$$\mathbf{DX} = \mathbf{F}(\mathbf{X}_{\text{LG}}, \mathbf{U}_{\text{LG}}) \quad (3.7)$$

where  $\mathbf{X}_{\text{LG}}$  is matrix of states at Legendre-Gauss node points,  $\mathbf{U}_{\text{LG}}$  is matrix of controls at Legendre-Gauss node points,  $\mathbf{X}$  is matrix of states at Legendre-Gauss node points and at initial point  $x_0$

This is then used as constraints to formulate a transcription method

$$\begin{aligned} \min \quad & \Phi(\mathbf{X}_{N+1}), \\ \text{s.t.} \quad & \mathbf{DX} = \mathbf{F}(\mathbf{X}_{\text{LG}}, \mathbf{U}_{\text{LG}}), \\ & \mathbf{X}_{N+1} = \mathbf{X}_0 + \mathbf{w}^T \mathbf{F}(\mathbf{X}_{\text{LG}}, \mathbf{U}_{\text{LG}}), \\ & \mathbf{X}_0 = \mathbf{x}_0. \end{aligned} \quad (3.8)$$

Where  $w$  are the Gauss quadrature weights. This is the differential form of Legendre-Gauss collocation method, integral form is described in [24].

### 3.4 Collocation at Legendre-Gauss-Radau points

This method is constructed in similar manner to previous, with difference that one endpoint is a collocation node. This is an advantage, as the resulting nonlinear programming formulation does not require the gauss quadrature rule for endpoint and it is smaller. This method approximates functions up degree  $2n - 2$  with machine precision, this is lower precision than Legendre-Gauss.

$$\begin{aligned} \min \quad & \Phi(\mathbf{X}_N) \\ \text{subject to} \quad & \mathbf{DX} = \mathbf{F}(\mathbf{X}^{\text{LGR}}, \mathbf{U}^{\text{LGR}}), \\ & \mathbf{X}_0 = \mathbf{x}_0. \end{aligned} \quad (3.9)$$

This is also a differential formulation.

### 3.5 Collocation at Legendre-Gauss-Lobatto points

The Gauss-Lobatto family includes both endpoints as collocation points, achieving a quadrature degree of exactness of  $2N - 3$ . It encompasses widely used schemes such as the trapezoidal and Hermite-Simpson methods. Legendre-Gauss-Lobatto method is formulated using Bucher table of Lobatto-III methods developed in [?]. Then constraints are formulated in classical RK way

### 3.6 Gauss methods as pseudospectral collocation

I have implemented Gauss-Legendre, Gauss-Lobatto, Gauss-Radau as global collocation methods, that is one polynomial approximating the entire trajectory. I have implemented them according to [23]

## 3.7 Gauss methods as $h$ -method

### 3.7.1 collocation integral form

integral form is straight forward, same as other Runge-Kutta Methods

### 3.7.2 collocation differential form

differential form requires to create differentiation matrix of the approximating polynomial, which contains the derivative of the approximating polynomial.

## 3.8 Mesh refinement methods

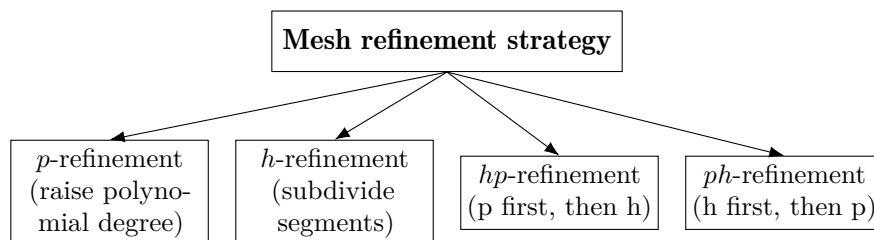
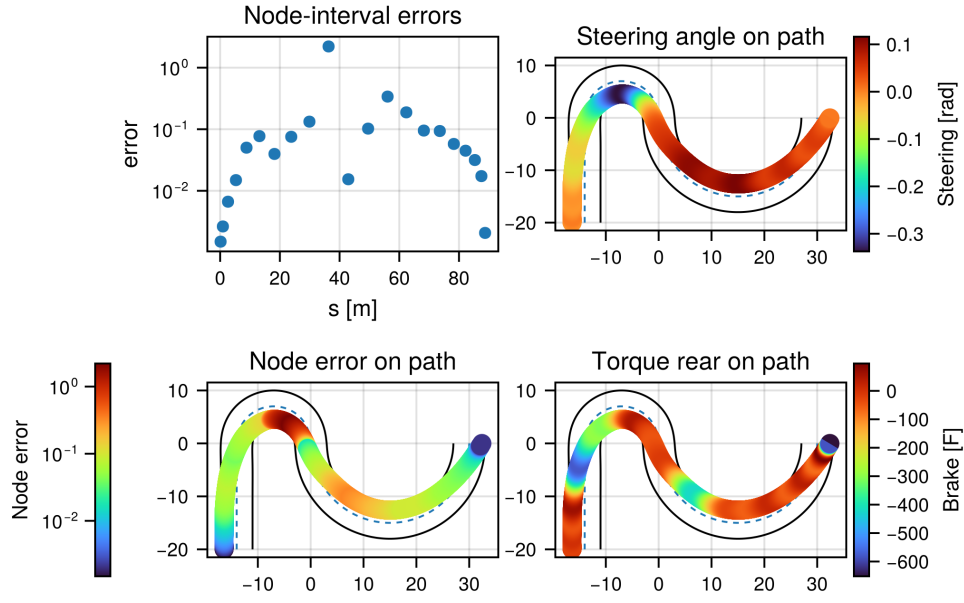


Figure 3.6: Mesh refinement [20, 48].

### 3.8.1 $p$ adaptive methods

$p$  adaptive methods work by increasing the order of polynomial, until the error is below user defined threshold. Figure 3.9 shows that error along track is not distributed equally. Error is localised around areas with large jump in controls. These areas need finer sampling.  $P$  methods cannot focus these specific points directly, they focus entire trajectory.

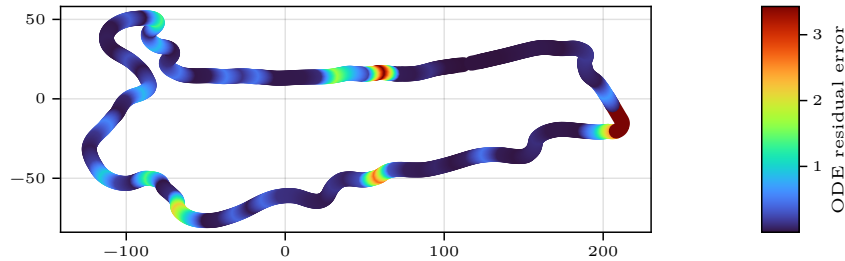
The errors are concentrated around the discontinuity of controls, where the behaviour of the vehicle is more complex/ requiring high order or finer approximation see Figure 3.9. Time optimal control of car around track has many discontinuities and therefore  $p$  methods are not a good fit for this use.



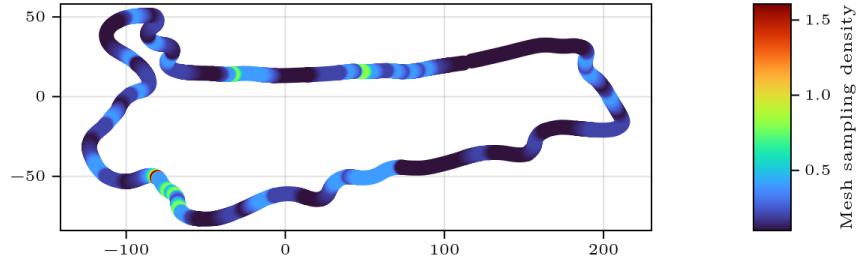
**Figure 3.7:** Node-interval ODE error for power-test case.

### 3.8.2 $h$ adaptive methods

$h$  methods are used when solution of optimal control problem is approximated by many intervals with low degree polynomial. These methods can decide to split segment into two smaller segments, they use different techniques to do this. Method described in [15] calculates error using Romberg quadrature with machine precision. Another method described in [31] uses techniques from essentially non oscillatory (ENO) interpolation



**Figure 3.8:** Node-interval ODE error for power-test case.



**Figure 3.9:** Node-interval ODE error for power-test case.

[27] to decide where to split and remove nodes.

### ■ 3.8.3 hp adaptive methods

hp adaptive methods are state of the art in mesh refinement. They are combination of increasing/decreasing mesh segments and increasing/decreasing order of approximating polynomial inside segments. One of the first hp-method and commonly cited is described in article [20] from University of Florida. They have produced numerous papers on hp- adaptive mesh refinement [9] [41]. Recent work on mesh refinement for functions with kinks [62]. They are the most complex for implementation and unfortunately I did not have the time to implement them.

## ■ 3.9 Adjoint methods

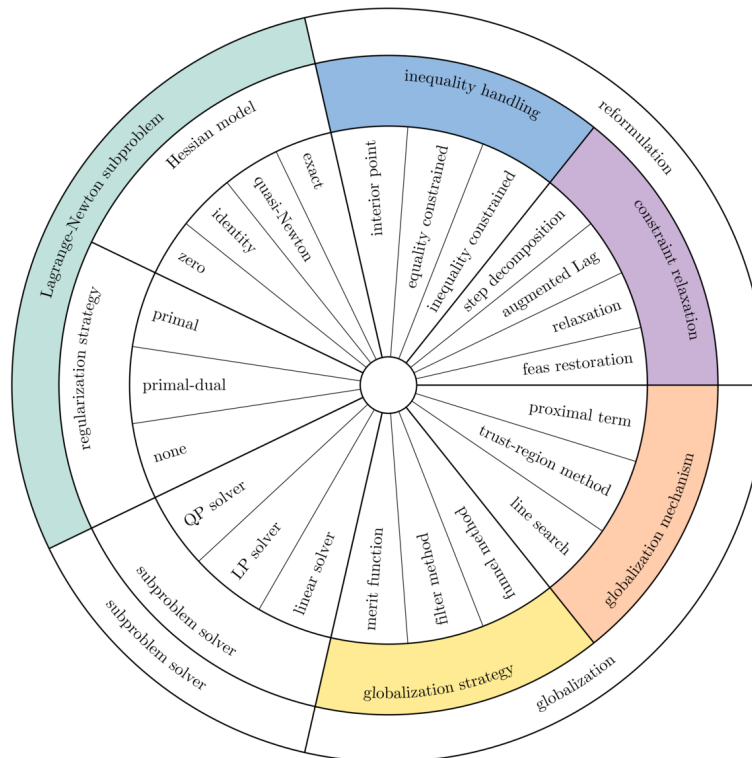
Adjoint methods allow for differentiating the solution of optimization problem. In this case it is function of time on states and controls along the trajectory and car parameters. This is very useful as it greatly simplifies sensitivity analysis. Traditional way of sensitivity analysis is to compute several runs with different vehicle parameters, but this allows to compute sensitivity of all parameters in one shot. I am using DiffOpt.jl [14], it is not exactly adjoint, I don't really understand very well how it works.



## Chapter 4

### Solvers for nonlinear programming

Once the objective function and constraints are created, the problem is passed to the NLP solver. Commonly used methods are Lagrange-Newton methods, these are sequential quadratic solvers and interior point solvers. These solvers are perhaps the most complex component used in this thesis, this chapter will cover them at high level, as full theory is beyond the scope. A comprehensive overview of Lagrange-Newton methods is in article [57], where the methods are broken down into key algorithmic components and unifying framework of solvers is presented. Figure 4.1 demonstrates the complexity of these methods.



**Figure 4.1:** Unifying framework: the “wheel of strategies”[57]

High level overview required for this thesis of Lagrange-Newton methods is in Figure 4.2.

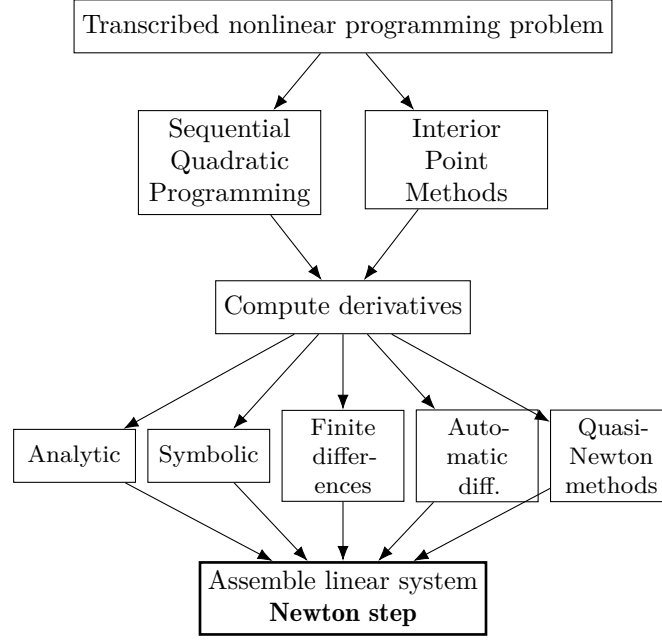


Figure 4.2: NLP solution flow

## 4.1 Nonlinear programming

General NLP problems can be written as:

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathbb{R}^n} \quad & f(\mathbf{x}) \\
 \text{s.t.} \quad & \mathbf{h}(\mathbf{x}) = 0 \\
 & \mathbf{g}(\mathbf{x}) \leq 0
 \end{aligned} \tag{4.1}$$

Here  $\mathbf{x} \in \mathbb{R}^n$  is the decision-variable vector,  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  is the scalar objective,  $\mathbf{h} : \mathbb{R}^n \rightarrow \mathbb{R}^m$  stacks the  $m$  equality constraints, and  $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^p$  stacks the  $p$  inequality constraints. In the lap-time setting  $\mathbf{x}$  holds the discretised states and controls,  $f$  is total lap time,  $\mathbf{h}$  enforces vehicle dynamics and boundary conditions, and  $\mathbf{g}$  enforces track limits, tire friction circle and actuator bounds.

### 4.1.1 KKT conditions

KKT conditions are the conditions of optimality, if satisfied the point is local optimum. alpha and omega of nonlinear programming, lord praise the KKT conditions.



$$\begin{aligned}
 \nabla f(\mathbf{x}) + \sum_{i=1}^m \lambda_i \nabla h_i(\mathbf{x}) + \sum_{i=1}^p \mu_i \nabla g_i(\mathbf{x}) &= 0 \\
 \mathbf{h}(\mathbf{x}) &= \mathbf{0} \\
 \mathbf{g}(\mathbf{x}) &\leq \mathbf{0} \\
 \mu_i &\geq 0, \quad i = 1, \dots, p \\
 \mu_i g_i(\mathbf{x}) &= 0, \quad i = 1, \dots, p
 \end{aligned} \tag{4.2}$$

## 4.2 Interior point methods: IPOPT, MadNLP

Interior point solvers or barrier solvers work by reformulating the NLP into a barrier problem at each step and solving a sequence of these barrier problems, with decreasing barrier parameter. Solver used in this thesis is IPOPT [63] and MadNLP[54].

### 4.2.1 Barrier problem formulation

There are multiple ways to transform NLP in to barrier problem, this section will briefly describe how IPOPT does it. IPOPT article [63] Describes how to transform generic NLP into form 4.3

$$\begin{aligned}
 \min_{\mathbf{x} \in \mathbb{R}^n} \quad & \phi_\mu(\mathbf{x}) := f(\mathbf{x}) - \mu \sum_{i=1}^n \ln(x_i^{(i)}) \\
 \text{s.t.} \quad & c(\mathbf{x}) = 0
 \end{aligned} \tag{4.3}$$

The Lagrangian of the barrier problem is defined as

$$\mathcal{L}(\mathbf{x}, \lambda, z) := f(\mathbf{x}) + c(\mathbf{x})^\top \lambda - z. \tag{4.4}$$

Then barrier problem is further transformed into set of linear equations, a KKT linear system, that computes the Newton step:

$$\begin{bmatrix} W_k & A_k & -I \\ A_k^\top & 0 & 0 \\ Z_k & 0 & X_k \end{bmatrix} \begin{bmatrix} d_x^k \\ d_\lambda^k \\ d_z^k \end{bmatrix} = - \begin{bmatrix} \nabla f(x_k) + A_k \lambda_k - z_k \\ c(x_k) \\ X_k Z_k e - \mu_j e \end{bmatrix}. \tag{4.5}$$

Here  $W_k = \nabla^2 \mathcal{L}(x_k, \lambda_k, z_k)$  is the Hessian of the Lagrangian,  $A_k$  is the Jacobian of the equality constraints  $c(x)$  and  $\nabla f(x_k)$  is the gradient of the objective. The unknowns  $d_x^k$ ,  $d_\lambda^k$  and  $d_z^k$  are the Newton steps in the primal variables, equality multipliers and bound multipliers;  $\mu_j$  is the barrier parameter, which is driven to zero across outer iterations.

There are numerous commercial and free algorithms to solve this system of linear equation. For smaller systems it is LU decomposition, but for larger sparse problems a dedicated solver is a must. IPOPT is most commonly used with MUMPS [10] solver which is free. There are several linear commercial solvers, Harwell Subroutine Library(HSL) offers ma27 ma87 ma97 [30].Panua

technologies offers PARDISO solver [53]. Harwell Subroutine Library solvers and PARDISO solver limited to 2 cores are available free for academic usage. These solvers run on CPU, Nvidia has linear solver CuDSS[47] which runs on GPU. Linear solvers are capable of multi threading on CPU, which needs to be explicitly turned on.

### 4.3 Sequential quadratic programming

Sequential quadratic programming is less commonly used for offline trajectory optimisation, usage of IPOPT prevails. celu sekciu prec

#### 4.4 Calculation of derivatives

A significant computational burden of Lagrange-Newton methods is that they require gradient of the objective function, Jacobian of the constraints and Hessian of the Lagrangian to compute every Newton step. Computing these derivatives can be time intensive. The most universal method is finite differences, which means evaluating the function around the current point and estimating the derivative, this works even for a black box model, but the derivatives are not exact. When the model is available, symbolic differentiation can be used, however, the full symbolic expression is often too large to be evaluated effectively due to repeated chain and product rules. Automatic differentiation aims to solve the problems of previous methods, by applying the chain rule to the elementary operations of a function. A comprehensive review of automatic differentiation is given in [19].

## 4.5 Scaling and matrix conditioning

Common pitfall of nonlinear programming is poor scaling. This happens when the solver encounters variables, that are orders of magnitude apart, for example motor force in thousands of newtons and heading angle in radians. Newton steps in IPOPT are theoretically scale invariant, but the rest of the algorithm is not. [63, p. 21]. With improper scaling, Lagrange multipliers in 4.5 may become large when active constraints are nearly linearly dependant and this causes numerical difficulties.

Similar problems happen when nonsmooth function is used in the model, like  $\text{abs}()$ ,  $\text{max}()$ , branching. These have to be replaced by smooth approximation or completely reformulated.

A related issue is constraints qualification. Linear independence constraint qualification requires gradients of the active constraints to be linearly independent. If it is violated, the Jacobian of the constraints is rank deficient

and linear solve may fail. This happens when for example one constraint is created twice.

## 4.6 Sparsity

Sparsity is a desirable property. Sparse Hessian and Jacobian matrices contain a large fraction of zero entries, which interior-point solvers exploit through specialized linear algebra. Sparsity in the constraints themselves is also important: individual constraints should involve only a small subset of the decision variables, rather than being formulated as deeply nested nonlinear expressions. Example of this is that state derivatives of vehicle at each point, can be fully determined from only 7 variables: accelerator pedal, brake pedal, steering angle, forward speed, lateral speed, yaw rate and heading. Using only these variables results in a deeply nested nonlinear expression in the constraints. Adding redundant decision variables like forces on all wheels can reduce the iterations needed and calculation time of derivatives. Counter intuitively this larger problem was faster to solve, this is elaborated on in [33, 34]. Certain solvers can heavily exploit the sparsity structure, such as FATROP[58]. In source [32] Joris Gillis a CasADi developer compares IPOPT to FATROP, with **FATROP being 33 times faster on a trajectory optimisation problem**. Also, [15] describes how sparsity structure helps with faster derivatives approximation with finite difference and Quasi-Newton methods.

## 4.7 Initialization

NLP solvers require initial guess for all decision variables, to start solving. For this problem initial guess is obtained driving the vehicle on given track. PD controller is used to keep the car on the centerline and P controller is used to keep velocity at 5m/s. In general, it is possible to initialize all variables at zero, but with large scale nonlinear programming this approach is not used. For this problem initialization with zeros leads to singularity in equation 2.20 and is not possible at all.

## 4.8 Global and local optimums

This optimization problem is generally non-convex, therefore any local optimum does not guarantee global optimum. Commonly used NLP solvers, like IPOPT guarantee only local optimums and not global ones. Good initialization is basis for finding a local optimum close to the global optimum [39, §5.1, p. 11].





## Part II

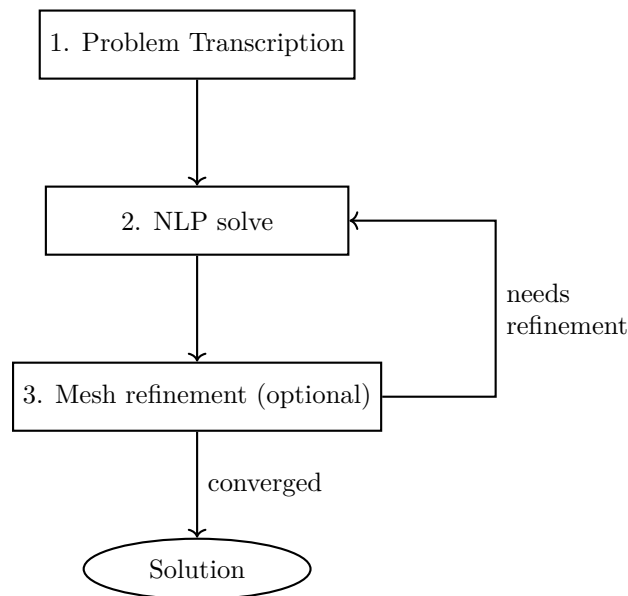
### Implementation of Optimal Control problem



## Chapter 5

### Frameworks for Optimal Control Problem Formulation

Process of solving optimal control problem has 3 parts. Problem definition, solving of nonlinear program, optional mesh refinement. This chapter describes frameworks in which the NLP problem can be created and solved. I have stated the theory, these are possible ways to implement the theory By framework I mean programming language + AD tool + mesh refining



**Figure 5.1:** Optimal control problem solving pipeline.

The consensus in optimal control community about tools for problem formulation is that:

There is no shortage of optimal control software based on pseudospectral approaches. There are a number of other optimal control libraries that tackle similar kinds of problems, such as OTIS4, GPOPS-II, and CASADI.

[21][?] This chapter is about tool that define the OCP, and supply derivatives of the problem, they cannot solve the problem by themselves.

[60] for automatic differentiation. This tool is commercial and closed source. Implementation in this thesis is heavily based on work of community from University of Florida.

## 5.1 CasADi

CasADi is an open-source tool for nonlinear optimization and algorithmic differentiation.[17][11]. interfaces with C++,Python,MATLAB. It provides a framework INE SLOVO to build the problem, and calculate the hessian and jacobian for nlp solver it does not have any mesh refinemnt algorithms. Commonly paired with IPOPT for nonliner programming, but is compatible with large variety of solvers.

## 5.2 dymos + OpenMDAO

Python has hp mesh refinement uses OpenMDAO [26] developed at NASA Glenn research center for evaluating aircraft concepts. Lobatto and radau collocation as default [21]. According to forum post from julia discourse, which may be biased towards julia it is slower compared to infiteOpt in julia [6]. But it has mesh refinement so that is good.

### 5.3 Acados

Acados is a modular open-source framework for fast embedded optimal control, developed at University of Freiburg [59]. It has C, Python, MATLAB + Simulink and Octave interface. This framework support generating C code. It is demonstrated by numerous real-world examples, such as drone racing with replanning, robot navigation in cluttered environment and many others [8].

## 5.4 GPOPS-II

GPOPS-II [49] is perhaps the most advanced tool for optimal control, it was developed at University of Florida by Michael Patterson and Anil V. Rao. Other tools allow defining the problem, but GPOPS-II is a level higher. It also includes advanced mesh refinement strategies, multiple phase OCP and automatic scaling. For solving the problem it uses IPOPT or SNOPT. It interfaces with MATLAB and C++, uses ADigator

## 5.5 Julia + JuMP/ infititeopt/ examomdels

ExaModels, JuMP



JuMP is the go to tool for algebraic modelling in julia, it is mature, versatile. Supports Linear programming, quadratic programming, mixed linear programming, Nonlinear programming. ExaModels [56] is a specialized algebraic modelling tool, it utilises Single instruction, multiple data methods, enabling derivative evaluation on GPU accelerators. Coupled with MadNLP solver, the entire computation of nonlinear optimisation can be run on a GPU.

Julia options

1. Julia + jump
2. Julia + jump + infinte opt
3. julia + ExaModels

Infite opt already has various collocation methods implemented. For learning purposes and in fear that in future down the road i would realise that infiteopt blocks all the stuff i want to do i decided to use jump, without infite opt.

No julia package has mesh refinement I chose julia + jump,



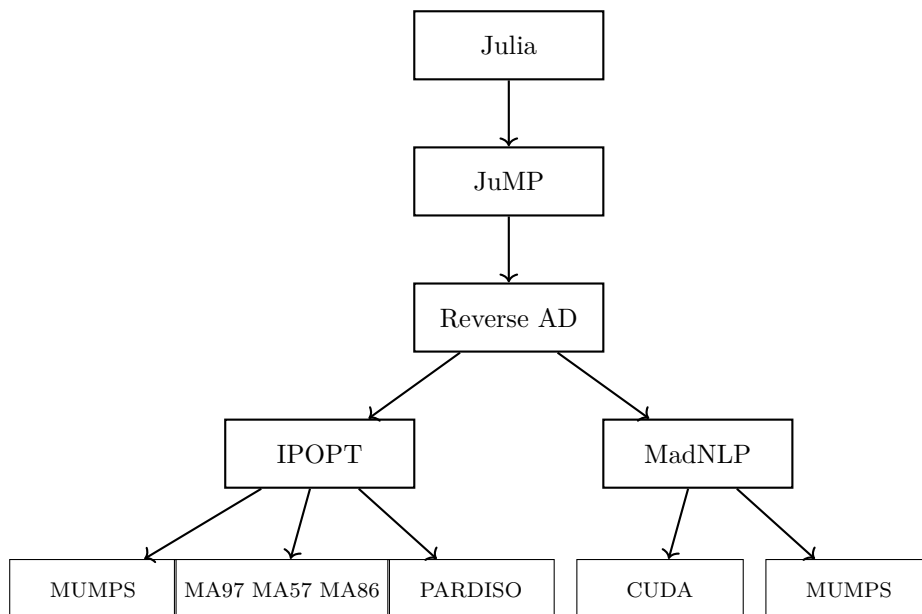
## Chapter 6

### My implementation, julia + JuMP

I have decided to use julia + JuMP, because I wanted to use free and open source software.

I have already done a lot in Python and julia seemed like new interesting thing to learn. Julia is supposed to be fast and high level language, bridging the gap between C++ and python. Julia packages offer a lot of ready to use mathematical tools developed by community. Julia has

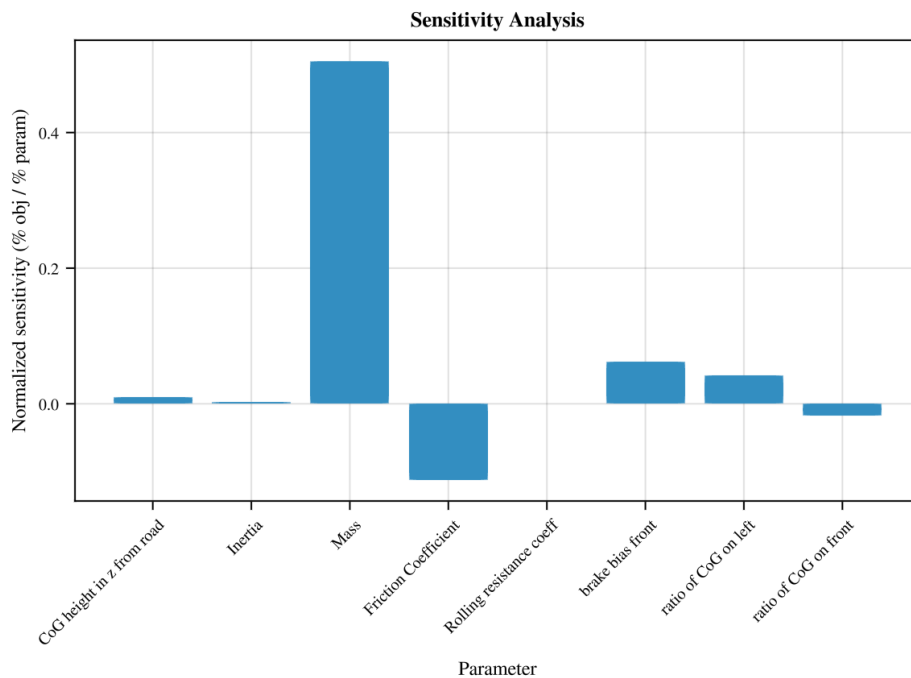
JuMP [42] (Julia for Mathematical Programming) is a modelling language and collection of optimization packages in Julia. JuMP makes it possible to create decision variables and constraints symbolically and uses automatic differentiation, to supply first and second derivative to solvers. Solver used in this thesis is IPOPT [63] and MadNLP[55][54]. MadNLP also allows utilising GPU for solver computation.



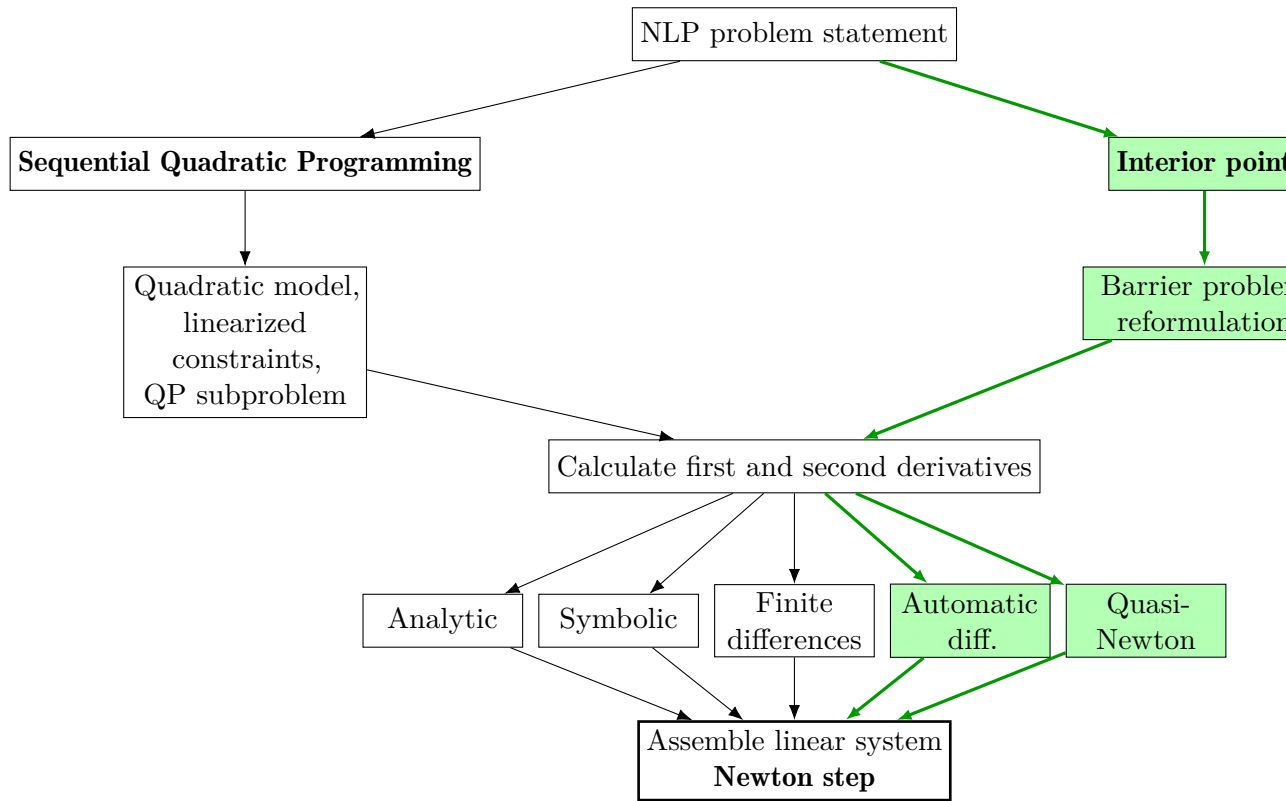
**Figure 6.1:** Optimization software stack

## 6.1 sensitivity Analysis

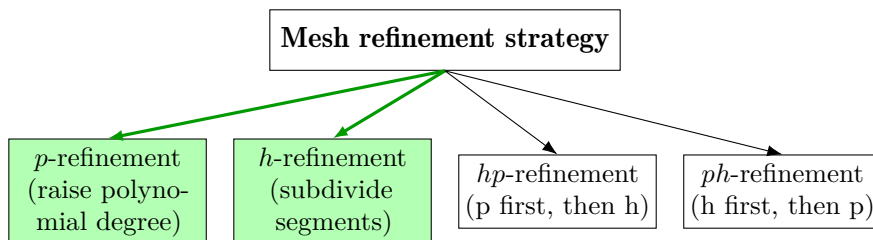
I have used DiffOpt.jl [14] for the sensitivity analysis. By adding vehicle parameters such as mass, lift coefficient, CoG position etc. into optimisation as decision variables fixed to a value by constraints. This adds them to the algebraic model, then the solution of minimum laptime can be differentiated by these parameters. AKO POMENOVAT TEN STROM CO JUMP VYTCORI TEN GRAF? algebraic model. Result of this is on figure 6.2



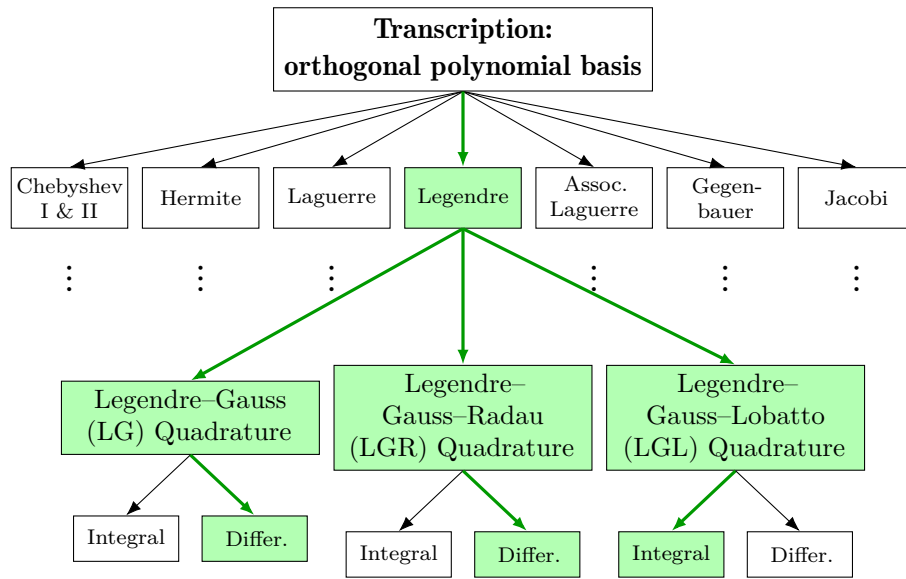
**Figure 6.2:** Node-interval ODE error for power-test case.



**Figure 6.3:** NLP solution flow with the methods used in this thesis highlighted in green



**Figure 6.4:** Mesh refinement layer with the strategies implemented in this thesis highlighted in green



**Figure 6.5:** Transcription layer with the methods used in this thesis highlighted in green



## **Part III**

### **results**





## Chapter 7

### comparison of methods

Performance of NLP solvers and frameworks is often tested on benchmark problems, like CUTEst [25] dataset. I have created my own dataset for this purpose. Tests were performed on Ryzen 9 9950X and MadNLP ran on Nvidia RTX 5070. Important caveat is that IPOPT has many (probably too many) options which can be adjusted, and that is true for all solvers. Some of the differences in this benchmark may be due to sub-optimal setup of the solver.

**Table 7.2:** Benchmark summary statistics by transcription variant.

| transcription | runs | ok | fail | rate     | mean_s   | med_s   | min_s  | max_s     |
|---------------|------|----|------|----------|----------|---------|--------|-----------|
| Legendre      | 14   | 13 | 1    | 92.8571  | 115.5586 | 25.9074 | 0.5774 | 1009.4537 |
| Lobatto IIIA  | 14   | 13 | 1    | 92.8571  | 66.4536  | 22.3981 | 0.2790 | 532.3975  |
| Radau         | 14   | 14 | 0    | 100.0000 | 71.6885  | 21.5821 | 0.4218 | 572.8931  |

**Table 7.4:** Benchmark results by track, vehicle model, and transcription variant.

| track       | car         | transcription | ok  | status             | time_s    | obj      |
|-------------|-------------|---------------|-----|--------------------|-----------|----------|
| FSCZ        | singletrack | Radau         | yes | solved             | 29.1453   | 50.2106  |
| FSCZ        | singletrack | Legendre      | yes | solved             | 49.7079   | 50.5890  |
| FSCZ        | singletrack | Lobatto IIIA  | yes | solved             | 28.0042   | 51.4710  |
| FSCZ        | twintrack   | Radau         | yes | solved             | 109.6182  | 57.3381  |
| FSCZ        | twintrack   | Legendre      | no  | locally infeasible | 381.2216  | NaN      |
| FSCZ        | twintrack   | Lobatto IIIA  | no  | locally infeasible | 224.8871  | NaN      |
| Berlin      | singletrack | Radau         | yes | solved             | 25.9621   | 79.4234  |
| Berlin      | singletrack | Legendre      | yes | solved             | 44.6565   | 79.4033  |
| Berlin      | singletrack | Lobatto IIIA  | yes | solved             | 39.2681   | 79.1299  |
| Berlin      | twintrack   | Radau         | yes | solved             | 67.7176   | 77.7439  |
| Berlin      | twintrack   | Legendre      | yes | solved             | 134.5996  | 77.7883  |
| Berlin      | twintrack   | Lobatto IIIA  | yes | solved             | 52.1696   | 77.8983  |
| Berlin      | formula     | Radau         | yes | solved             | 65.8456   | 87.9477  |
| Berlin      | formula     | Legendre      | yes | solved             | 96.1020   | 88.3235  |
| Berlin      | formula     | Lobatto IIIA  | yes | solved             | 76.7914   | 87.9482  |
| Berlin      | bus         | Radau         | yes | solved             | 572.8931  | 106.9827 |
| Berlin      | bus         | Legendre      | yes | solved             | 1009.4537 | 106.6620 |
| Berlin      | bus         | Lobatto IIIA  | yes | solved             | 532.3975  | 105.8631 |
| FSG         | singletrack | Radau         | yes | solved             | 17.2021   | 58.2463  |
| FSG         | singletrack | Legendre      | yes | solved             | 25.9074   | 58.1609  |
| FSG         | singletrack | Lobatto IIIA  | yes | solved             | 22.3981   | 56.8773  |
| FSG         | twintrack   | Radau         | yes | solved             | 86.4238   | 69.3149  |
| FSG         | twintrack   | Legendre      | yes | solved             | 109.9323  | 70.3348  |
| FSG         | twintrack   | Lobatto IIIA  | yes | solved             | 95.6528   | 69.8878  |
| DoubleTurn  | singletrack | Radau         | yes | solved             | 0.4218    | 5.7087   |
| DoubleTurn  | singletrack | Legendre      | yes | solved             | 0.5774    | 5.6794   |
| DoubleTurn  | singletrack | Lobatto IIIA  | yes | solved             | 0.2790    | 6.0247   |
| DoubleTurn  | twintrack   | Radau         | yes | solved             | 2.1042    | 6.5283   |
| DoubleTurn  | twintrack   | Legendre      | yes | solved             | 1.2756    | 6.5835   |
| DoubleTurn  | twintrack   | Lobatto IIIA  | yes | solved             | 1.5196    | 7.0205   |
| DoubleTurn  | bus         | Radau         | yes | solved             | 2.2998    | 8.9075   |
| DoubleTurn  | bus         | Legendre      | yes | solved             | 1.9180    | 8.9864   |
| DoubleTurn  | bus         | Lobatto IIIA  | yes | solved             | 1.8477    | 8.8676   |
| FigureEight | singletrack | Radau         | yes | solved             | 1.5903    | 14.9127  |
| FigureEight | singletrack | Legendre      | yes | solved             | 1.9768    | 14.9162  |
| FigureEight | singletrack | Lobatto IIIA  | yes | solved             | 1.3477    | 14.8296  |
| FigureEight | twintrack   | Radau         | yes | solved             | 12.5541   | 16.5690  |
| FigureEight | twintrack   | Legendre      | yes | solved             | 20.7391   | 16.5736  |
| FigureEight | twintrack   | Lobatto IIIA  | yes | solved             | 9.2105    | 16.5319  |
| FigureEight | bus         | Radau         | yes | solved             | 9.8610    | 22.8420  |
| FigureEight | bus         | Legendre      | yes | solved             | 5.4157    | 22.7966  |
| FigureEight | bus         | Lobatto IIIA  | yes | solved             | 3.0111    | 22.3442  |

**Table 7.6:** Linear solver benchmark summary statistics.

| linear solver | runs | ok | fail | rate     | mean_s   | med_s   | min_s  | max_s     |
|---------------|------|----|------|----------|----------|---------|--------|-----------|
| ma27          | 14   | 14 | 0    | 100.0000 | 97.1562  | 12.1841 | 0.2777 | 964.2969  |
| ma57          | 14   | 14 | 0    | 100.0000 | 93.3249  | 12.6211 | 0.2897 | 895.8892  |
| ma97          | 14   | 14 | 0    | 100.0000 | 107.8802 | 12.1752 | 0.2689 | 1099.4816 |
| mumps         | 14   | 14 | 0    | 100.0000 | 69.6295  | 12.6470 | 0.2803 | 569.8588  |

**Table 7.8:** Linear solver benchmark results by track, vehicle model, and transcription variant.

| track       | car         | linear solver | ok  | status | time_s    | obj      |
|-------------|-------------|---------------|-----|--------|-----------|----------|
| FSCZ        | singletrack | mumps         | yes | solved | 11.4861   | 50.2106  |
| FSCZ        | singletrack | ma57          | yes | solved | 11.2823   | 50.2106  |
| FSCZ        | singletrack | ma27          | yes | solved | 11.2823   | 50.2106  |
| FSCZ        | singletrack | ma97          | yes | solved | 11.2701   | 50.2106  |
| FSCZ        | twintrack   | mumps         | yes | solved | 108.9638  | 57.3381  |
| FSCZ        | twintrack   | ma57          | yes | solved | 91.0531   | 57.3381  |
| FSCZ        | twintrack   | ma27          | yes | solved | 108.9439  | 57.2790  |
| FSCZ        | twintrack   | ma97          | yes | solved | 92.4024   | 57.3381  |
| Berlin      | singletrack | mumps         | yes | solved | 26.6890   | 79.4234  |
| Berlin      | singletrack | ma57          | yes | solved | 24.6851   | 79.4234  |
| Berlin      | singletrack | ma27          | yes | solved | 25.5493   | 79.4234  |
| Berlin      | singletrack | ma97          | yes | solved | 25.6775   | 79.4234  |
| Berlin      | twintrack   | mumps         | yes | solved | 67.2587   | 77.7439  |
| Berlin      | twintrack   | ma57          | yes | solved | 58.9130   | 77.7439  |
| Berlin      | twintrack   | ma27          | yes | solved | 51.7979   | 77.7439  |
| Berlin      | twintrack   | ma97          | yes | solved | 55.8040   | 77.7439  |
| Berlin      | formula     | mumps         | yes | solved | 64.5359   | 87.9477  |
| Berlin      | formula     | ma57          | yes | solved | 62.7170   | 87.9477  |
| Berlin      | formula     | ma27          | yes | solved | 60.3951   | 87.9477  |
| Berlin      | formula     | ma97          | yes | solved | 73.0484   | 87.9477  |
| Berlin      | bus         | mumps         | yes | solved | 569.8588  | 106.9827 |
| Berlin      | bus         | ma57          | yes | solved | 895.8892  | 106.9795 |
| Berlin      | bus         | ma27          | yes | solved | 964.2969  | 106.9815 |
| Berlin      | bus         | ma97          | yes | solved | 1099.4816 | 106.6410 |
| FSG         | singletrack | mumps         | yes | solved | 12.7246   | 58.2463  |
| FSG         | singletrack | ma57          | yes | solved | 13.9599   | 58.2463  |
| FSG         | singletrack | ma27          | yes | solved | 13.0858   | 58.2463  |
| FSG         | singletrack | ma97          | yes | solved | 13.0492   | 58.2463  |
| FSG         | twintrack   | mumps         | yes | solved | 85.2451   | 69.3149  |
| FSG         | twintrack   | ma57          | yes | solved | 123.3800  | 69.3471  |
| FSG         | twintrack   | ma27          | yes | solved | 100.8860  | 69.3140  |
| FSG         | twintrack   | ma97          | yes | solved | 115.3881  | 69.3140  |
| DoubleTurn  | singletrack | mumps         | yes | solved | 0.2803    | 5.7087   |
| DoubleTurn  | singletrack | ma57          | yes | solved | 0.2897    | 5.7087   |
| DoubleTurn  | singletrack | ma27          | yes | solved | 0.2777    | 5.7087   |
| DoubleTurn  | singletrack | ma97          | yes | solved | 0.2689    | 5.7087   |
| DoubleTurn  | twintrack   | mumps         | yes | solved | 2.0312    | 6.5283   |
| DoubleTurn  | twintrack   | ma57          | yes | solved | 2.8587    | 6.5283   |
| DoubleTurn  | twintrack   | ma27          | yes | solved | 1.8017    | 6.5283   |
| DoubleTurn  | twintrack   | ma97          | yes | solved | 1.9207    | 6.5283   |
| DoubleTurn  | bus         | mumps         | yes | solved | 2.2462    | 8.9075   |
| DoubleTurn  | bus         | ma57          | yes | solved | 1.5365    | 8.9075   |
| DoubleTurn  | bus         | ma27          | yes | solved | 1.3301    | 8.9075   |
| DoubleTurn  | bus         | ma97          | yes | solved | 1.3733    | 8.9075   |
| FigureEight | singletrack | mumps         | yes | solved | 0.8838    | 14.9127  |
| FigureEight | singletrack | ma57          | yes | solved | 0.8841    | 14.9127  |
| FigureEight | singletrack | ma27          | yes | solved | 0.9921    | 14.9127  |
| FigureEight | singletrack | ma97          | yes | solved | 0.9997    | 14.9127  |
| FigureEight | twintrack   | mumps         | yes | solved | 12.5693   | 16.5690  |
| FigureEight | twintrack   | ma57          | yes | solved | 11.0038   | 16.5690  |
| FigureEight | twintrack   | ma27          | yes | solved | 11.1527   | 16.5690  |
| FigureEight | twintrack   | ma97          | yes | solved | 11.3013   | 16.5690  |
| FigureEight | bus         | mumps         | yes | solved | 10.0401   | 22.8420  |
| FigureEight | bus         | ma57          | yes | solved | 8.0956    | 22.8420  |
| FigureEight | bus         | ma27          | yes | solved | 8.3957    | 22.8420  |
| FigureEight | bus         | ma97          | yes | solved | 8.3372    | 22.8420  |

**Table 7.10:** NLP solver / backend benchmark summary statistics (IPOPT-MUMPS, MadNLP-CPU, MadNLP-GPU).

| NLP solver  | runs | ok | fail | rate     | mean_s   | med_s   | min_s  | max_s     |
|-------------|------|----|------|----------|----------|---------|--------|-----------|
| ipopt-mumps | 14   | 14 | 0    | 100.0000 | 126.2746 | 12.3629 | 0.2810 | 1347.3147 |
| madnlp-cpu  | 14   | 14 | 0    | 100.0000 | 126.0898 | 17.1029 | 0.2816 | 1370.6252 |
| madnlp-gpu  | 14   | 14 | 0    | 100.0000 | 155.0950 | 20.4672 | 0.4655 | 1733.7308 |

**Table 7.12:** NLP solver / backend benchmark results by track, vehicle model, and transcription variant.

| track       | car         | NLP solver  | ok  | status                | time_s    | obj      |
|-------------|-------------|-------------|-----|-----------------------|-----------|----------|
| FSCZ        | singletrack | ipopt-mumps | yes | solved                | 11.5065   | 50.2106  |
| FSCZ        | singletrack | madnlp-cpu  | yes | solved                | 18.3244   | 50.2106  |
| FSCZ        | singletrack | madnlp-gpu  | yes | solved                | 70.2315   | 50.2098  |
| FSCZ        | twintrack   | ipopt-mumps | yes | solved                | 109.2145  | 57.3381  |
| FSCZ        | twintrack   | madnlp-cpu  | yes | solved                | 103.5061  | 57.2490  |
| FSCZ        | twintrack   | madnlp-gpu  | yes | solved                | 92.1590   | 57.2521  |
| Berlin      | singletrack | ipopt-mumps | yes | solved                | 25.6021   | 79.4234  |
| Berlin      | singletrack | madnlp-cpu  | yes | solved                | 18.5674   | 79.4234  |
| Berlin      | singletrack | madnlp-gpu  | yes | solved                | 26.4438   | 79.4214  |
| Berlin      | twintrack   | ipopt-mumps | yes | solved                | 87.9163   | 77.7417  |
| Berlin      | twintrack   | madnlp-cpu  | yes | solved                | 59.0629   | 77.7442  |
| Berlin      | twintrack   | madnlp-gpu  | yes | solved                | 45.7640   | 77.7421  |
| Berlin      | formula     | ipopt-mumps | yes | solved                | 64.0543   | 87.9477  |
| Berlin      | formula     | madnlp-cpu  | yes | solved                | 60.2524   | 87.9477  |
| Berlin      | formula     | madnlp-gpu  | yes | almost locally solved | 77.4451   | 87.9191  |
| Berlin      | bus         | ipopt-mumps | yes | solved                | 1347.3147 | 107.2120 |
| Berlin      | bus         | madnlp-cpu  | yes | solved                | 1370.6252 | 106.6728 |
| Berlin      | bus         | madnlp-gpu  | yes | solved                | 1733.7308 | 106.8060 |
| FSG         | singletrack | ipopt-mumps | yes | solved                | 12.3721   | 58.2463  |
| FSG         | singletrack | madnlp-cpu  | yes | solved                | 13.4788   | 58.2463  |
| FSG         | singletrack | madnlp-gpu  | yes | solved                | 13.4315   | 58.2455  |
| FSG         | twintrack   | ipopt-mumps | yes | solved                | 83.2545   | 69.3149  |
| FSG         | twintrack   | madnlp-cpu  | yes | solved                | 90.3133   | 69.4039  |
| FSG         | twintrack   | madnlp-gpu  | yes | solved                | 85.6687   | 69.3864  |
| DoubleTurn  | singletrack | ipopt-mumps | yes | solved                | 0.2810    | 5.7087   |
| DoubleTurn  | singletrack | madnlp-cpu  | yes | solved                | 0.2816    | 5.7087   |
| DoubleTurn  | singletrack | madnlp-gpu  | yes | solved                | 0.4655    | 5.7086   |
| DoubleTurn  | twintrack   | ipopt-mumps | yes | solved                | 1.9413    | 6.5283   |
| DoubleTurn  | twintrack   | madnlp-cpu  | yes | solved                | 2.6051    | 6.5463   |
| DoubleTurn  | twintrack   | madnlp-gpu  | yes | solved                | 2.3967    | 6.5282   |
| DoubleTurn  | bus         | ipopt-mumps | yes | solved                | 1.4252    | 8.9075   |
| DoubleTurn  | bus         | madnlp-cpu  | yes | solved                | 1.3106    | 8.9075   |
| DoubleTurn  | bus         | madnlp-gpu  | yes | solved                | 1.9968    | 8.8951   |
| FigureEight | singletrack | ipopt-mumps | yes | solved                | 0.8828    | 14.9127  |
| FigureEight | singletrack | madnlp-cpu  | yes | solved                | 0.8406    | 14.9127  |
| FigureEight | singletrack | madnlp-gpu  | yes | solved                | 1.2380    | 14.9124  |
| FigureEight | twintrack   | ipopt-mumps | yes | solved                | 12.3536   | 16.5690  |
| FigureEight | twintrack   | madnlp-cpu  | yes | solved                | 15.8815   | 16.5658  |
| FigureEight | twintrack   | madnlp-gpu  | yes | solved                | 5.8682    | 16.5652  |
| FigureEight | bus         | ipopt-mumps | yes | solved                | 9.7260    | 22.8420  |
| FigureEight | bus         | madnlp-cpu  | yes | solved                | 10.2077   | 22.7936  |
| FigureEight | bus         | madnlp-gpu  | yes | solved                | 14.4905   | 22.7981  |

**Table 7.14:** Hessian approximation benchmark summary statistics (exact vs. limited-memory quasi-Newton).

| hessian mode | runs | ok | fail | rate     | mean_s   | med_s   | min_s  | max_s     |
|--------------|------|----|------|----------|----------|---------|--------|-----------|
| ipopt-exact  | 14   | 14 | 0    | 100.0000 | 124.7120 | 13.5311 | 0.2756 | 1317.5593 |
| ipopt-lbfgs  | 14   | 7  | 7    | 50.0000  | 12.8208  | 14.0032 | 0.2919 | 29.6997   |

**Table 7.16:** Hessian approximation benchmark results by track, vehicle model, and transcription variant.

| track       | car         | hessian mode | ok  | status                | time_s    | obj      |
|-------------|-------------|--------------|-----|-----------------------|-----------|----------|
| FSCZ        | singletrack | ipopt-exact  | yes | solved                | 11.5460   | 50.2106  |
| FSCZ        | singletrack | ipopt-lbfgs  | yes | solved                | 14.0032   | 50.2106  |
| FSCZ        | twintrack   | ipopt-exact  | yes | solved                | 107.1464  | 57.3381  |
| FSCZ        | twintrack   | ipopt-lbfgs  | no  | iteration limit       | 797.5930  | NaN      |
| Berlin      | singletrack | ipopt-exact  | yes | solved                | 27.4134   | 79.4234  |
| Berlin      | singletrack | ipopt-lbfgs  | yes | solved                | 29.6997   | 79.4234  |
| Berlin      | twintrack   | ipopt-exact  | yes | solved                | 89.4514   | 77.7417  |
| Berlin      | twintrack   | ipopt-lbfgs  | no  | iteration limit       | 2547.2935 | NaN      |
| Berlin      | formula     | ipopt-exact  | yes | solved                | 66.6850   | 87.9477  |
| Berlin      | formula     | ipopt-lbfgs  | no  | iteration limit       | 796.0425  | NaN      |
| Berlin      | bus         | ipopt-exact  | yes | solved                | 1317.5593 | 107.2120 |
| Berlin      | bus         | ipopt-lbfgs  | no  | iteration limit       | 695.7941  | NaN      |
| FSG         | singletrack | ipopt-exact  | yes | solved                | 14.5996   | 58.2463  |
| FSG         | singletrack | ipopt-lbfgs  | yes | solved                | 15.0913   | 58.2463  |
| FSG         | twintrack   | ipopt-exact  | yes | solved                | 84.5573   | 69.3149  |
| FSG         | twintrack   | ipopt-lbfgs  | no  | iteration limit       | 1387.7969 | NaN      |
| DoubleTurn  | singletrack | ipopt-exact  | yes | solved                | 0.2756    | 5.7087   |
| DoubleTurn  | singletrack | ipopt-lbfgs  | yes | solved                | 0.2919    | 5.7087   |
| DoubleTurn  | twintrack   | ipopt-exact  | yes | solved                | 1.9790    | 6.5283   |
| DoubleTurn  | twintrack   | ipopt-lbfgs  | yes | almost locally solved | 21.8729   | 6.5412   |
| DoubleTurn  | bus         | ipopt-exact  | yes | solved                | 1.4282    | 8.9075   |
| DoubleTurn  | bus         | ipopt-lbfgs  | yes | solved                | 7.7314    | 8.9075   |
| FigureEight | singletrack | ipopt-exact  | yes | solved                | 0.8776    | 14.9127  |
| FigureEight | singletrack | ipopt-lbfgs  | yes | solved                | 1.0555    | 14.9127  |
| FigureEight | twintrack   | ipopt-exact  | yes | solved                | 12.4626   | 16.5690  |
| FigureEight | twintrack   | ipopt-lbfgs  | no  | iteration limit       | 521.8558  | NaN      |
| FigureEight | bus         | ipopt-exact  | yes | solved                | 9.9866    | 22.8420  |
| FigureEight | bus         | ipopt-lbfgs  | no  | iteration limit       | 73.1088   | NaN      |



## Chapter 8

### Conclusion

All goals were achieved. One thing which was not a goal but, would be super useful if i managed to do a validation on real world data. Validation of sensitivity analysis. I wanted to do it for a long time but there was not enough of it.

Julia was interesting language to learn, it offers blazing fast execution, but that comes with caveats. Debugging is way more complicated. When working on a smaller project, the debugging is not such a pain, but when the project is bigger debugging is not user friendly. Debugging in python and MATLAB is way easier. Julias selling point is that it is just in time compiled language, rivalling C. That is true BUT you have to know what you are doing, if you dont declare types in functions it is as slow as python.

At the begging of this project I wanted a more complex model, I wanted to go deeper with complexity than my bachelor thesis. That is add pitch, roll and full model of suspension into the simulation. Also this simulation does not account for gearboxes with discrete stages, and no combustion engine.

Believing 100% this tool and designing car according to it would most likely lead to undriveable vehicle, as there is no force pushing for stability driveability in NLP [50] deals also with this. The optimal driving created by NLP is unachievable by real driver, as the vehicle is on its limit and one mistake woul overload tires and cause failure.

Track model is also heavily simplified, neglecting slope and bank of the track.







## Appendices



## Appendix A

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